RF Transceiver Module of a Low Cost Ground Penetrating Radar for Landmine Detection

B .Tech. Project

Submitted in partial fulfillment of the requirements of the degree of

Bachelor of Technology
in
Electrical Engineering

By

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Declaration

We hereby declare that the project entitled "RF Transceiver Module of a Low Cost Ground Penetrating Radar for Landmine Detection" submitted in partial fulfillment for the award of the degree of Bachelor of Technology in 'Electrical Engineering' completed under the supervision of Prof. Rinkee Chopra, Assistant Professor - Electrical Engineering department, IIT Indore is an authentic work.

Further, we declare that we have not submitted this work for the award of any other degree elsewhere.

We declare that this written submission represents my ideas in our own words, and where other's ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misinterpreted or fabricated or falsified any idea/data/fact/source in our submission. We understand that any violation of the above will be cause for disciplinary action by the institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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Preface

This report on **“ RF Transceiver Module of a Low Cost Ground Penetrating Radar for Landmine Detection ”** is prepared under the guidance of **Prof. Rinkee Chopra**.

Through this report, we have attempted to present a comprehensive design and development process of a low-cost RF transceiver module for Ground Penetrating Radar (GPR) aimed at landmine detection. The work aims to build a practical, efficient, and affordable solution based on FMCW radar principles. We've tried to present the technical concepts in a clear and straightforward way so the content stays both accessible and informative. To make things easier to follow, we've also added schematic diagrams, circuit layouts, and measured results, giving the report a detailed and visually helpful structure.

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B. Tech. IV year

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B. Tech. Project Approval Certificate

This is to certify that the dissertation titled **" RF Transceiver Module of a Low Cost Ground Penetrating Radar for Landmine Detection "** submitted by **Satya Sai Praneeth Karri** (Roll No. 220002042) and **Mohibahemad Ansari** (Roll No. 220002011) is approved for the award of degree of **Bachelor of Technology** in **Electrical Engineering.**

BTP Supervisor: 2/12/25

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Abstract

Presently, the Indian Security Force is heavily dependent on hand-held metal detectors to detect the land mines. These detectors have severe depth limitation in detection and they cannot give pseudo image or depth information about the target. Also, non-metal land mines cannot be detected by these metal detectors.

This project focuses on the design and implementation RF Transceiver for ground penetrating radar (GPR). The initial literature survey covers various GPR types, architectures and design parameters. Considering the limitations of existing systems, a low cost, homodyne, Frequency Modulated Continuous Wave (FMCW) radar centered at 700 MHz has been developed.

Individual RF modules such as VCO, power amplifies, LNA, band pass filters, demodulators were designed and tested extensively. The test results on the raw reflected data confirms the working of this system and that this can be extended into a high-performance GPR platform for reliable landmine detection.

Keywords – **Ground Penetrating Radar, FMCW, RF Transceiver, Landmine Detection, I-Q Demodulator, Voltage Controlled Oscillator, Low Nosie Amplifier**

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List of Abbreviations

RF	Radio Frequency
GPR	Ground Penetrating Radar
FMCW	Frequency Modulated Continuous Wave
PCB	Printed Circuit Board
IC	Integrated Circuit
EM	Electromagnetic (Waves)
GHz / MHz / kHz	Gigahertz / Megahertz / Kilohertz
GSM	Global System for Mobile Communications
CDMA	Code Division Multiple Access
SNR	Signal-to-Noise Ratio
DAC	Digital to Analog Converter
S-parameters	Scattering Parameters
dBm	Decibel-milliwatts
VCO	Voltage Controlled Oscillator
TWG	Triangular Wave Generator
SDR	Software Defined Radio
LNA	Low Noise Amplifier
CPWG	Co-planar Waveguide with Ground Plane
BPF	Band-Pass Filter
IQ / I-Q	In-phase and Quadrature
ADC	Analog to Digital Converter

Chapter 1

Introduction

1.1 Introduction

Landmines remain a serious global threat, causing thousands of deaths every year. Estimates suggest that more than 100 million unexploded mines of various types are still buried across nearly 70 countries. At the current pace and with existing technology, clearing all of them could take several centuries if no new mines were added [1]. This highlights the urgent need for more effective techniques for detecting and clearing minefields.

Landmines are generally classified into two categories: anti-personnel (AP) and anti-tank (AT). Anti-tank mines are larger and contain a greater amount of explosive material, whereas anti-personnel mines are compact, often around 10 cm in diameter, and may be housed in plastic or other low-metal materials. Their small size and minimal metal content make AP mines particularly difficult to detect using conventional sensing methods [2].

Developing a complete landmine detection and localization system involves multiple stages [3]: (i) Detection—determining whether any buried object is present in the area of interest; (ii) Recognition—identifying whether the detected object is a landmine or just clutter such as stones, wood, or debris; (iii) Localization—estimating the exact position of suspected mines for safe excavation or neutralization.

Various sensing technologies have been explored for landmine detection. Some approaches use electromagnetic induction (EMI) sensors to identify small metallic components. However, since many modern landmines contain little to no metal, these methods often fall short. As a result, Ground Penetrating Radar (GPR) has gained significant attention as a more reliable option for detecting both metallic and non-metallic mines [4]. GPR is a subsurface sensing technique that sends high-frequency radio waves into the ground and captures the reflections caused by contrasts in electrical properties between buried objects and surroundings. Its ability to detect both metallic and non-metallic materials makes it a widely used method for locating landmines and other hidden hazards [5].

1.2 Motivation for BTP and Problem statement

1.2.1 Motivation

Conventional mine-detection methods, such as metal detectors and mechanical probing, face critical limitations. Metal detectors struggle with mines that have minimal or non-metallic content, and they suffer high false-alarm rates due to metallic clutter in the subsurface environment [6]. Mechanical or manual probing is slow, dangerous, and often unfeasible in difficult terrain or under vegetation cover. In contrast, ground-penetrating radar (GPR) has become a strong option for non-intrusive subsurface sensing. It works by sending electromagnetic waves into the ground and analyzing the reflected signals to identify anomalies created by buried objects whose dielectric properties or geometry differ from the surrounding material [7]. Different GPR techniques, such as FMCW and stepped-frequency methods, offer their own benefits in terms of radar front-end design and system integration. For example, FMCW radar allows continuous transmission and reception of the waveform, which supports a more compact design, better signal-to-noise performance, and smoother hardware integration [8].When applied to landmine detection, an FMCW GPR system can deliver high-resolution subsurface imaging and distinguish mines from clutter by leveraging differences in spectral, dielectric, and temporal signatures. This helps reduce false alarms and improves overall detection reliability [9]. However, using such systems in real minefields comes with major engineering challenges. The subsurface environment is extremely variable—factors like soil moisture, mineral content, surface roughness, and vegetation all influence how electromagnetic waves propagate and reflect. Mines may be shallow, deeply buried, metallic or plastic, clustered with clutter (rocks, roots, pipes), and often in complex terrains [10]. Moreover, for practical deployment, the radar front-end must be compact, low-power (especially for mobile or unmanned platforms), and must interface with digital signal-processing chains and hardware modules reliably [11].

1.2.2 Problem Statement

Ground Penetrating Radar (GPR) based on Frequency Modulated Continuous Wave (FMCW) operation can overcome several of these limitations by sending radio waves into the ground and analyzing their reflections to identify buried objects. For such systems, the transceiver is the most critical block as it decides how cleanly the signal is transmitted and how effectively the weak echoes are captured. Designing this RF front end with low noise, wide bandwidth, and stable frequency sweep is still a challenging task, especially when the system has to be compact and power-efficient for field use. The focus of this project is to design and verify an RF transceiver module that can generate and process FMCW signals suitable for GPR-based landmine detection.The module we developed is designed to balance sensitivity, range resolution, and hardware simplicity, providing a solid and reliable foundation for future radar enhancements.

1.3 Organization of the Report

This report is divided into six main chapters, followed by an appendix and references.

Chapter one provides the project background, explains the motivation behind choosing this topic, and clearly states the problem addressed in this work.

Chapter two reviews the existing literature on Ground Penetrating Radar (GPR), covering its working principles, different system types, and the key design parameters that affect overall performance.

Chapter three describes the design and implementation of the RF transceiver module developed for the FMCW GPR in detail. It describes the overall system architecture, transmitter and receiver chain designs, and the various RF components such as the VCO, amplifiers, filters, and I-Q demodulator used in the design.

Chapter four explains the process of integrating all the designed modules into a complete GPR system.

Chapter five covers the experimental setup used for testing the developed modules and presents the results obtained for different target materials in air and soil conditions.

Chapter six summarizes the major conclusions drawn from the project and suggests possible directions for future improvement, including deeper penetration capability and the use of advanced signal-processing or machine-learning techniques.

The last section lists all references consulted during the course of this work, and the appendix includes the STM32 microcontroller and Python codes used for data acquisition and visualization.

Chapter 2

Literature Review

2.1 Principle of Operation of GPR

Ground Penetrating Radar (GPR) works on the basic idea of how electromagnetic (EM) waves interact with different materials below the surface. The transmitting antenna sends a continuous or pulsed EM wave towards the ground. When this wave encounters a change in the electrical properties of the subsurface, such as a boundary between two materials with different dielectric constants, a portion of the wave energy is reflected back, while the rest continues to travel deeper. These reflections are then captured by the receiving antenna, as illustrated in Fig. 2.1.

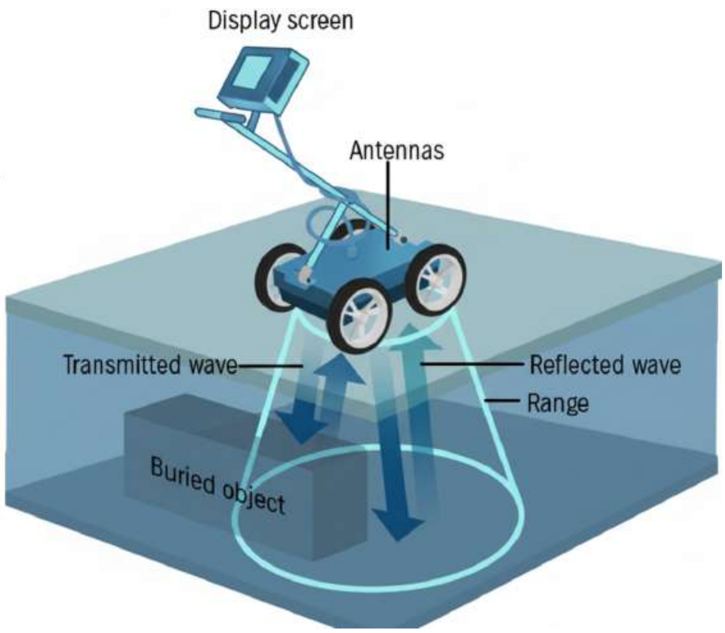


Fig. 2.1: Principal of Operation of GPR

The strength and phase of the received signals depend on the electromagnetic characteristics of the materials present underground, mainly their relative permittivity, electrical conductivity, and attenuation constant. Materials with high dielectric contrast, such as metal objects or voids in soil, produce stronger reflections, while materials with similar dielectric properties cause weaker or more diffused echoes. By measuring the time delay between the transmitted and received signals, the system can estimate the depth and nature of the buried object or layer.

At its core, GPR works like an imaging tool for what lies beneath the surface. The radar continuously scans the target area, and the returned signals are processed to generate profiles that show how the reflected energy varies with time and position. These profiles reveal anomalies such as mines, pipes, or rocks that differ from the surrounding soil. The system's performance strongly depends on the operating frequency: higher frequencies offer better resolution but limited depth, while lower frequencies penetrate deeper at the cost of detail. With well-designed antennas and suitable signal processing, GPR provides a non-invasive and reliable way to detect and analyze buried objects.

2.2 Types of GPR Systems

Ground Penetrating Radar (GPR) systems are mainly categorized based on how they transmit and process electromagnetic signals. The three most common types are, impulse (time-domain) GPR, stepped-frequency (SF) GPR systems and frequency-modulated continuous wave (FMCW) GPR. Each of these operates on a different signal principle and has its own advantages depending on the application.

2.2.1 Impulse GPR

Impulse GPR transmits very short-duration electromagnetic pulses into the ground using a broadband antenna. When these pulses encounter a boundary between materials with different dielectric properties, a portion of the energy is reflected back. By measuring the time delay between the transmitted and received pulses, the depth and location of buried objects can be estimated. The received signals are then processed to generate a time-domain profile of the subsurface.

The advantages of using impulse GPR are

1. It provides high temporal resolution, enabling precise depth estimation.
2. It has simple interpretation since data is collected directly in the time domain.
3. It is suitable for deep subsurface imaging, depending on the operating frequency.

Disadvantages of using impulse GPR is,

1. Requires high-speed data acquisition and wideband antennas, increasing system complexity and cost.

2. The short pulse energy is spread across a wide frequency band, often leading to low signal-to-noise ratio (SNR).

3. High power consumption due to the need for fast transient circuits and amplifiers.

2.2.2 Stepped Frequency GPR

Stepped-frequency GPR works by sending out a series of continuous sinusoidal signals, each at a different, discrete frequency within the chosen bandwidth. For every frequency step, the system records the amplitude and phase of the reflected signal. Once the sweep is complete, these measurements are converted into the time domain using an inverse Fourier transform to generate a depth profile. In this way, the method creates an effective broadband response from a set of narrowband transmissions.

The advantages of using Stepped Frequency GPR is

1. Offers excellent control over bandwidth and frequency range, leading to high depth resolution.

2. The use of narrowband signals improves dynamic range and measurement accuracy.

3. Provides flexibility in frequency selection, which helps reduce interference and optimize performance for different soil types.

The disadvantages of using Stepped Frequency GPR are

1. Slower data acquisition since multiple frequency steps are required for one scan.

2. More complex signal processing is needed to reconstruct time-domain responses.

3. Sensitive to phase errors between frequency steps, which can distort the final image.

2.2.3 FMCW GPR

In FMCW GPR, a continuous electromagnetic wave is transmitted whose frequency increases (or decreases) linearly over time, often called a “chirp.” The reflected wave from underground targets is combined (mixed) with a portion of the transmitted signal to produce a beat frequency that corresponds to the round-trip delay. The distance to the target is then determined from this frequency difference, and by sweeping over a range of frequencies, a complete depth profile can be reconstructed.

The advantages of using it FMCW GPR are
i) Provides high range resolution with lower power requirements.
ii) Improved signal-to-noise ratio (SNR) due to coherent signal processing.
iii) The use of continuous wave transmission allows simpler hardware and smaller antennas compared to impulse GPR.
iv) Well-suited for real-time detection applications like landmine and object detection.

In this project, we have chosen the FMCW GPR architecture due to its superior SNR, hardware simplicity, and suitability for real-time detection. This makes it the most efficient choice for our landmine detection system.

2.2.3.1 Theory and Mathematical Background of FMCW Demodulation

Frequency Modulated Continuous Wave (FMCW) radar operates by continuously transmitting a frequency–modulated signal (usually a linear chirp) and mixing the received echo with a portion of the transmitted signal to obtain a beat signal. The frequency of this beat signal is proportional to the round-trip delay and, hence, to the target distance.

Principle of FMCW Signal Generation

The transmitted signal of an FMCW radar can be expressed as

$$s_t(t) = A_t \cos\left[2\pi \left(f_c t + \frac{kt^2}{2}\right)\right], \tag{2.1}$$

where A_t is the transmit amplitude, f_c is the carrier frequency, and $k = \frac{B}{T_m}$ is the chirp rate with sweep bandwidth B and modulation period T_m.

The received signal reflected from a target at a distance R is a delayed and attenuated version of the transmitted signal:

$$s_r(t) = A_r \cos\left[2\pi \left(f_c(t-\tau) + \frac{k(t-\tau)^2}{2}\right)\right], \tag{2.2}$$

where $\tau = \frac{2R}{c}$ is the propagation delay and c is the speed of light.

Beat Frequency and Range Estimation

When the transmitted and received signals are mixed, the resulting difference component (after low-pass filtering) gives the beat frequency:

$$f_b = k\tau = \frac{2kR}{c} = \frac{2BR}{cT_m}. \tag{2.3}$$

Hence, the target range can be calculated as

$$R = \frac{cT_m f_b}{2B}. \tag{2.4}$$

For triangular or sawtooth modulation, separate up and down-sweeps help distinguish stationary and moving targets by comparing frequency differences, enabling Doppler estimation.

I/Q Demodulation and Complex Baseband Representation

An FMCW receiver uses an I/Q demodulator (such as the ADL5380) to obtain in-phase (I) and quadrature (Q) components of the received signal. These are obtained by mixing the received RF signal with local oscillator (LO) signals that are 90° apart:

$$I(t) = s_r(t) \times \cos(2\pi f_c t), \tag{2.5}$$

$$Q(t) = s_r(t) \times \sin(2\pi f_c t). \tag{2.6}$$

After low-pass filtering to remove the high-frequency terms, the baseband outputs become

$$I(t) = A\cos(\phi(t)), \tag{2.7}$$

$$Q(t) = A\sin(\phi(t)), \tag{2.8}$$

where $\phi(t) = 2\pi f_b t + \phi_0$ represents the instantaneous phase of the beat signal. Combining I and Q forms the complex baseband signal:

$$s_{bb}(t) = I(t) + jQ(t) = Ae^{j\phi(t)}. \tag{2.9}$$

This complex representation preserves both amplitude and phase information, allowing precise determination of target distance and motion through FFT processing.

Extraction of Target Information

The frequency of the baseband signal corresponds to target range, while its phase variation over time contains Doppler information. Performing an FFT on $s_{bb}(t)$ provides peaks at beat frequencies proportional to different target ranges. For multiple reflections, the output spectrum will contain multiple peaks corresponding to various targets.

In summary, I/Q demodulation converts the high-frequency FMCW echo into a low-frequency complex baseband signal, from which both range and velocity of the target can be accurately extracted.

2.3 Design Parameters and Challenges

Various system parameters such as choice of frequency, bandwidth, resolution, penetration depth, system power, system performance factor and dynamic range have been briefly discussed in this section. These parameters are extremely useful from the designer aspect [12].

2.3.1 Selection of Frequency

The performance of GPR is primarily evaluated by its depth of penetration and resolution.The selection of optimal frequency of GPR is a complex task, as it influences depth of penetration, spatial resolution and clutter rejection.

2.3.1.1 Depth of Penetration

As stated above, there exists a tradeoff between the depth of penetration and spatial resolution. Higher frequency systems ensure better resolution, and hence millimeter waves are used for subsurface profiling. Similar to the depth of penetration, an empirical relation has been derived for frequency, which determines the resolution of GPR [13] , and is given by Equation (2.10).

$$F_R \geq \frac{75}{x\sqrt{\varepsilon_0}} \tag{2.10}$$

where x is the distance in meters. The above parameters put a constraint on the center frequency, which is expressed as:

$$F_R < F < \min(F_D, F_C) \tag{2.11}$$

Table **??** provides a rough estimate for selecting the frequency required to achieve a particular depth of penetration [13] . These values are obtained practically under the best possible conditions (at optimum power levels).

Center Frequency (MHz)	**Depth (m)**
1000	0.5
500	1.0
200	2.0
100	7.0
50	10.0
25	30.0
10	50.0

Table 2.1: Variation of Depth of Penetration with Frequency

2.3.1.2 Spatial Resolution

Spatial resolution indicates how well the GPR system can separate two objects that lie close to each other in depth. It mainly depends on the operating frequency and the bandwidth of the transmitted signal. Higher frequencies produce shorter wavelengths, which helps in identifying smaller features with better clarity. However, this improvement comes at the cost of reduced penetration in highly lossy soil. In general, wider bandwidth and a higher center frequency lead to finer resolution, while the material's dielectric constant and signal attenuation limit the achievable detail.

2.3.1.3 Clutter Rejection

Clutter refers to unwanted echoes generated by rough surfaces, small stones, vegetation roots, or any irregularities in the soil. These unwanted reflections can obscure the actual target and degrade the clarity of the radar image. A GPR system's ability to suppress this clutter depends on factors like the operating frequency, antenna design, and the processing methods applied. Lower frequencies typically produce smoother responses and are less affected by small scatterers, while higher frequencies tend to capture more clutter. Methods such as background subtraction, filtering, time-gating, and careful antenna placement can improve the signal-to-clutter ratio, making the subsurface response easier to interpret.

Chapter 3

Design and Implementation of GPR Transceiver Module

3.1 System Architecture and Design Considerations

3.1.1 GPR Architecture

The frequency modulated continuous wave (FMCW) design was selected for its accurate range measurements and efficient signal processing [14]. The homodyne architecture was chosen to keep the system simple and cost-effective [15]. The general block diagram of a GPR using homodyne architecture is given in Fig. 3.1.

3.1.1.1 Transmitter

In the transmitter chain, a stable Frequency Modulated Continuous Wave (FMCW) signal is generated by using Voltage Controlled Oscillator (VCO). A Triangular Wave Generator (TWG) can be used for the modulation of frequency over the required band. Alternatively, a microcontroller with DAC can also be used to generate the same. An RF amplifier is typically placed after the VCO to boost the signal power, enhancing the depth of detection. A power divider routes part of the signal to the IQ demodulator as the Local Oscillator (LO) input and to the antenna for transmission.

3.1.1.2 Receiver

In the receiver chain, reflected signals from the subsurface objects are captured by the receiving antenna and are then amplified using a Low Noise Amplifier (LNA). The unwanted signal components are removed using a Bandpass Filter (BPF), after which the filtered signal is mixed with the LO in an IQ demodulator to produce the I and Q base-band outputs. These signals are then conditioned and digitized using an ADC, after which a microcontroller processes them to extract information about the target surface.

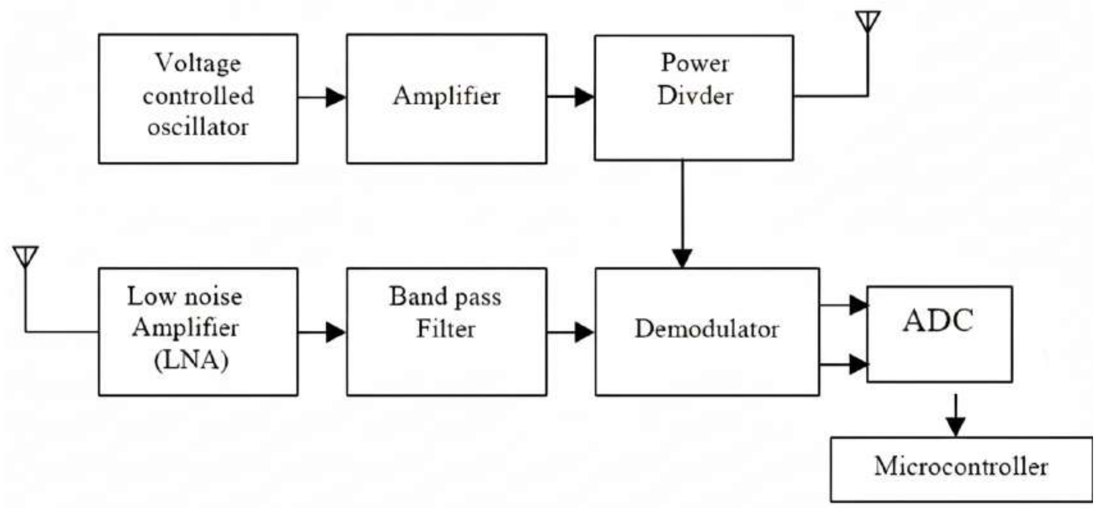


Fig. 3.1: Block diagram of a homodyne GPR system

3.1.2 Selection of Operating Frequency Band

The operating frequency significantly influences the penetration depth, spatial resolution, and clutter rejection of a GPR system. In order to achieve optimal performance, it is necessary to balance these interdependent design parameters during the complex process of choosing the center frequency.

As was previously mentioned, the resolution attained is directly proportional to the frequency of electromagnetic waves in a medium, whereas their skin depth is inversely proportional to it. A center frequency of 700 MHz is chosen to guarantee an operational depth of more than 60 cm while preserving good resolution. To avoid interfering with the nearby 800–900 MHz bands, which are designated for GSM and CDMA telecommunication services, the bandwidth is limited to about 180 MHz [16].

3.2 Transmitter Chain Design and Implementation

This section discusses the various RF modules that are part of the transmitter chain, including their design, implementation, and test results. These modules consist of the Voltage Controlled Oscillator (VCO), Triangular Wave Generator (TWG), RF Amplifier, and Power Divider. There is no need for any voltage conversion circuitry as all active modules are made to run on a 5 V supply. For stable operation, low signal distortion, and high-resolution subsurface imaging, these blocks must be carefully designed and integrated.

3.2.1 Voltage Controlled Oscillator (VCO)

The Voltage-Controlled Oscillator (VCO) generates the required transmission frequencies and serves as a critical component in the transmitter chain. An oscillator circuit typically consists of an amplifier that provides the necessary gain and a feedback network that determines the frequency of oscillation. In VCO, variation in frequency is typically achieved by varying one of the passive components in the feedback network. Depending on the components used in the feedback network, different types of VCOs can be designed, such as LC, RC, or varactor-tuned oscillators, each offering its own range of frequency tuning and performance characteristics.

3.2.1.1 Specifications of VCO

The target detection in FMCW GPR is mainly based on the variations in phase of the reflected signal and the frequency difference between current local oscillator signal and received signal called beat frequency. To get these accurately, VCO should exhibit low phase noise and also high frequency stability. Also, voltage pushing (change in frequency with supply voltage) should be kept as low as possible to reduce phase noise.

VCO shall provide a frequency sweep range of at least 180 MHz, centered at 700 MHz. For improved tuning stability, it is ideal to set the VCO's center frequency at a control voltage of VDD/2, ensuring that the overall tuning voltage waveform remains symmetric. The VCO should operate linearly across its entire range to ensure a consistent and flat output power throughout.

3.2.1.2 Design and PCB Layout

For achieving good phase noise and frequency stability, wideband colpitt's oscillator design was used [17]. In colpitt's VCO, capacitors in the feedback path are varied to achieve change in oscillation frequency. In the VCO circuit given in Fig. 3.2, variable capacitors are realized by varactor diodes BBY5702 [18] for their precise electronic control of frequency. For providing gain, BFP520 bipolar RF transistor [19], known for its low noise figure, high gain and excellent linearity, was used. Inductor and capacitor values are carefully chosen to get an oscillation frequency of around 700 MHz at a tuning voltage of 2.5 V.

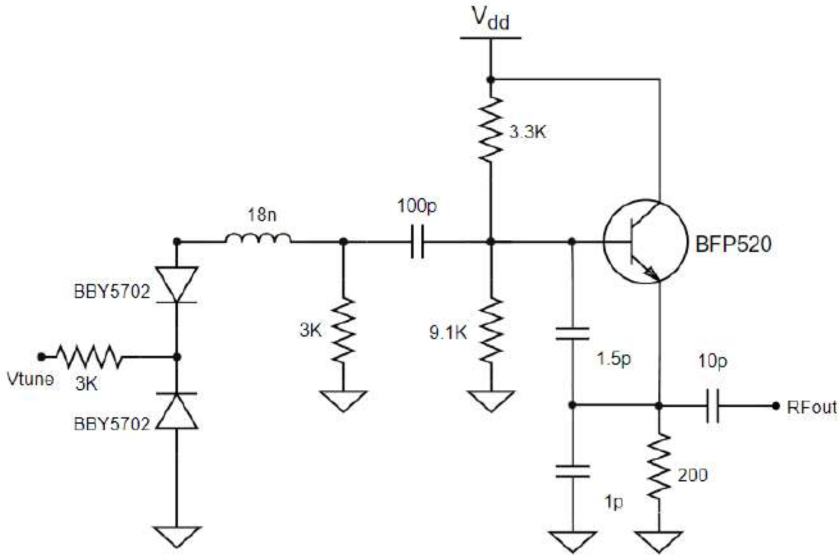
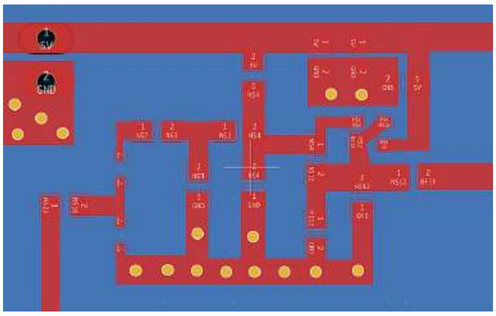
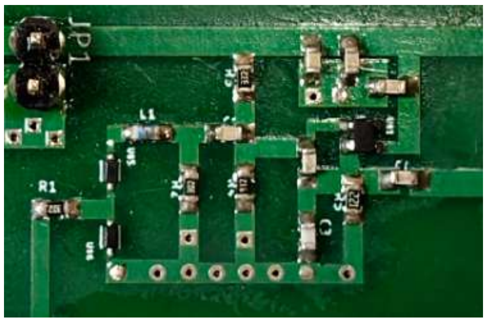


Fig. 3.2: Circuit schematic of the VCO

As the circuit operates in the microwave frequencies, properly designed PCB is essential for keeping parasitic inductance and capacitance to minimum. For designing the PCBs, KiCad software was chosen for its open-source and user-friendly interface and large library support. PCB for the VCO was designed while ensuring proper trace widths for impedance matching. Multiple decoupling capacitors were installed between the power line and ground to ensure the grounding of any noise present in the supply, thereby achieving a stable supply voltage that is critical for maintaining low phase noise. Multiple vias were used for better grounding. PCB layout is shown in Fig. 3.3(a) and the final PCB for testing is shown in Fig. 3.3(b).



(a) PCB Layout for VCO



(b) Populated PCB of VCO

Fig. 3.3: PCB Layout and Final PCB for the VCO

3.2.1.3 Measured Results and Analysis

The experimental setup consists of a Scientech 4077 - DC power supply and Rhode & Schwartz FPH.06 spectrum analyzer. The output spectrum of VCO observed at a supply voltage of 5 V and a tuning voltage of 2.5 V is shown in Fig. 3.4. It can be seen that the output frequency is at 700 MHz and the output power is approximately -2 dBm.

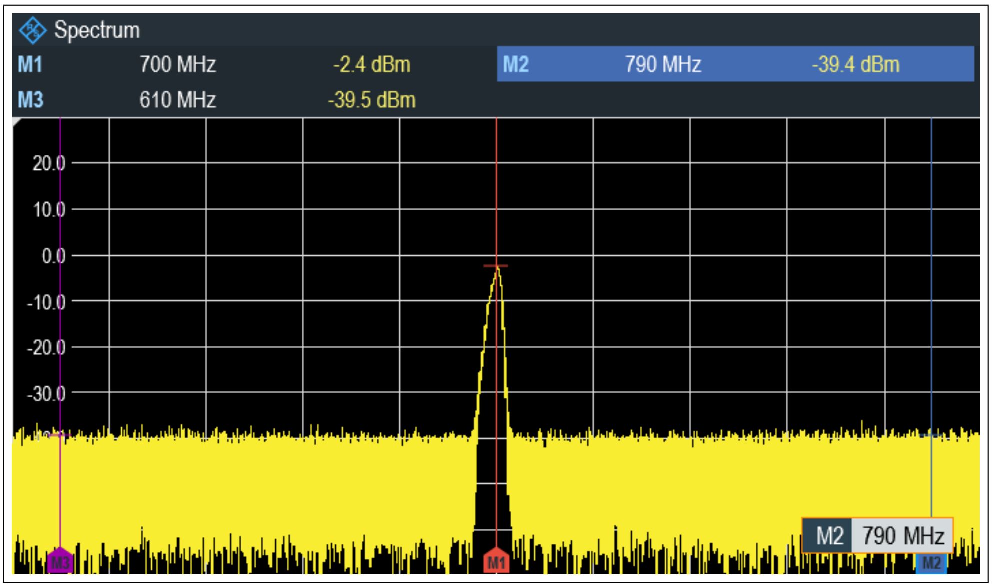


Fig. 3.4: Output Spectrum of VCO at tuning voltage of 2.5 V

As the supply voltage was fixed at 5 V and multiple decoupling capacitors were used between the power line and ground, frequency of oscillation was reduced to be mainly a function of the input tuning voltage. The frequency of oscillation was observed while changing the tuning voltage from 0.5 to 4.5 V to determine the voltage values for a required bandwidth of around 180 MHz centered at 700 MHz i.e. from 610 MHz to 790 MHz. The corresponding values for 610 MHz and 790 MHz were found to be 1 V and 4 V, resulting in the modulation sensitivity of 60 MHz/V . The output power of the VCO remained consistent within a 2 dBm variation across the entire frequency band, which is considered acceptable. Conclusively, the desired frequency band of 610 MHz to 790 MHz was achieved by applying a triangular tuning voltage ranging from 1 V to 4 V, generated using the TWG discussed in the following section.

3.2.2 Triangular Wave Generator (TWG)

The Triangular Wave Generator (TWG) is mainly used to generate a triangular waveform that is used for the modulation of VCO to generate a continuous frequency band in the required band of interest. Alternatively, a microcontroller with a Digital-to-Analog Converter (DAC) can be used for this purpose.

3.2.2.1 Design Specifications

To get a continuous band of frequencies from 610 MHz to 790 MHz, tuning voltage should linearly vary from 1 V to 4 V as discussed before. So, a triangular wave varying from 1 V to 4 V is required. The frequency of the triangular wave determines the chirp rate (rate of change of the VCO output frequency) and consequently influences the beat frequency. These aspects are discussed in detail in the IQ Demodulator section. Since a higher beat frequency requires a correspondingly higher ADC sampling rate for accurate acquisition, a moderate triangular wave frequency of around 10 kHz was selected.

3.2.2.2 Implementation

The circuit used for the generation of triangular wave is given in Fig. 3.5. AD8039 dual operational amplifier IC [20] was used where the first stage serves as an astable multivibrator to generate a square waveform and the second stage acts as an integrator to generate a triangular waveform. The corresponding PCB layout and final populated PCB are shown in Fig. 3.6(a) and Fig. 3.6(b) respectively. Multiple decoupling capacitors were connected between the power line and ground to effectively filter out high-frequency noise from the supply. Multiple vias were used for better grounding.

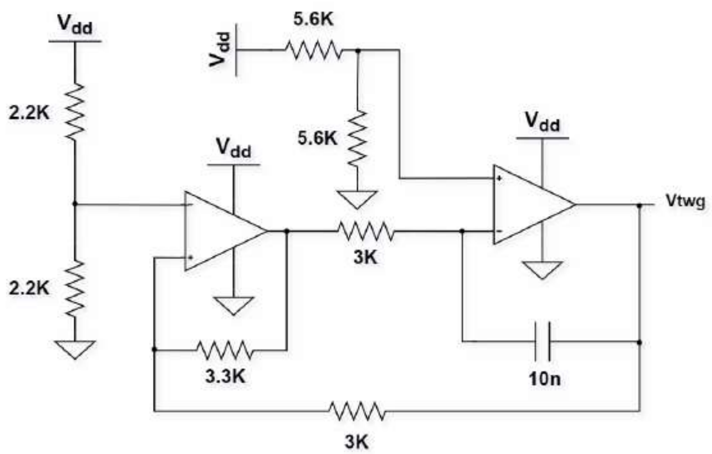
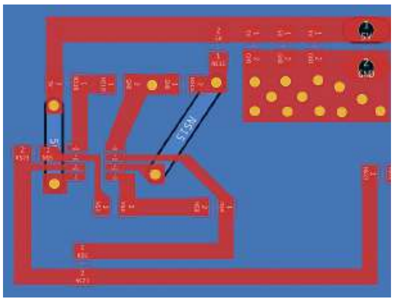
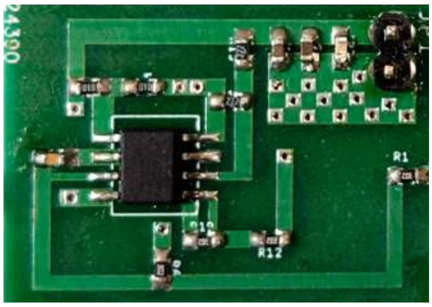


Fig. 3.5: Circuit schematic of TWG



(a) PCB layout for TWG

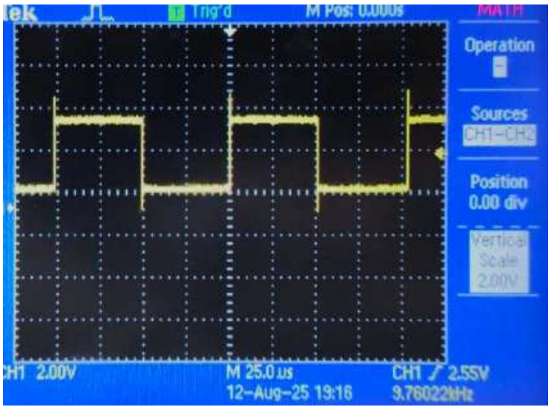


(b) Populated PCB of TWG

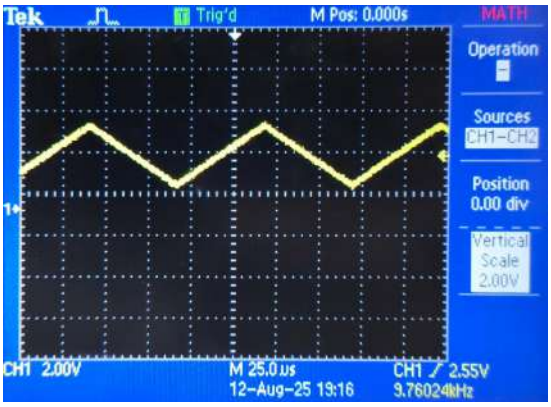
Fig. 3.6: PCB Layout and Populated PCB of the TWG

3.2.2.3 Measured Results

The experimental setup consists of a Scientech 4077 - DC power supply and Tektronix TDS1012C-EDU Digital Oscilloscope. The outputs of astable multi-vibrator and the integrator sections of TWG as seen in the oscilloscope are given in the Fig. 3.7(a) and Fig. 3.7(b) respectively. The observed frequency is 9.75 kHz and peak values of triangular wave are around 1 V and 4 V . These values are closely matching with the specifications required and were good to proceed with as the tuning voltage for the VCO.



(a) Output of astable multivibrator section of TWG



(b) Final output of TWG

Fig. 3.7: Square wave and final output waveforms of the TWG

3.2.2.4 Measured Results with VCO

This output of TWG is given as the tuning voltage for the VCO and the resulting output spectrum of VCO seen in the spectrum analyzer is given in Fig. 3.8. The output spectrum is observed to be centered around 700 MHz, and the bandwidth is nearly equal to 180 MHz as required. The output power level is approximately around –3 to -2 dBm and remains nearly flat throughout the frequency range from 610 MHz to 790 MHz. This level is adequate for subsequent amplification required for transmission.

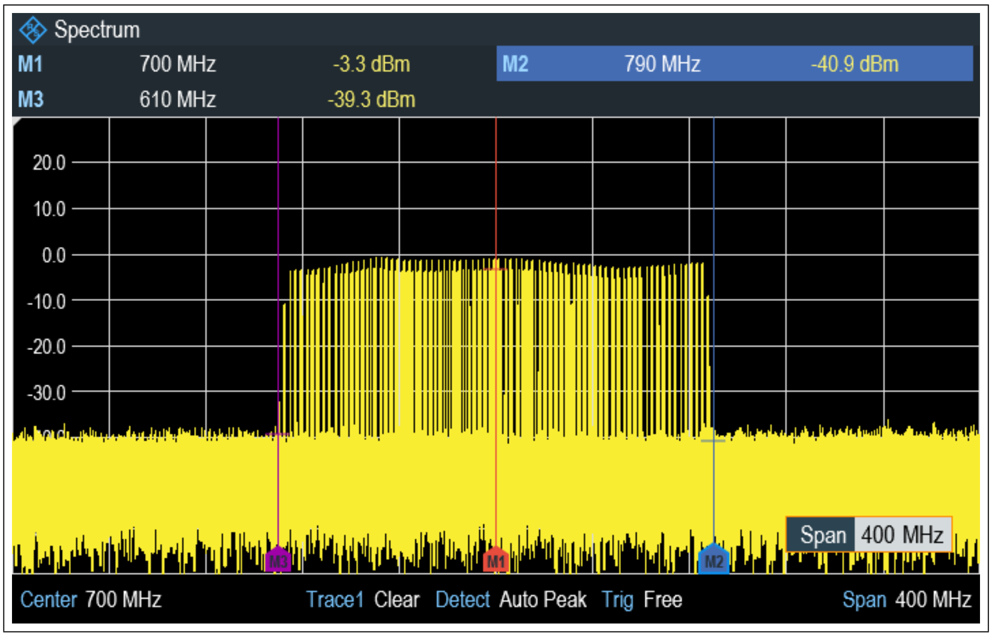


Fig. 3.8: Output of VCO with TWG's output as tuning voltage

3.2.3 RF Amplifier

In general, the output signal from the VCO has low power and requires amplification to achieve a greater detection range. This can be accomplished using an RF amplifier.

3.2.3.1 Design and Implementation

The design of amplifier circuits generally consists of designing biasing and matching networks to obtain the optimal gain from the selected IC. ADL5602 RF amplifier IC [21] has been used for this purpose due to its internal matching circuitry and good gain of 20 dB with an Output 1 dB Compression Point (P1dB) of 20 dBm. This eliminates the process of designing the external matching circuit for the amplifier, reducing the number of components required for its implementation shown in Fig. 3.9.

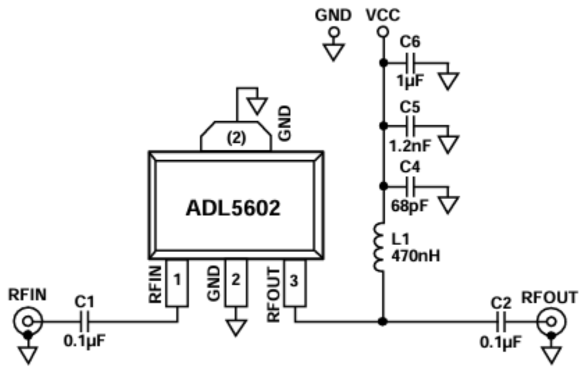
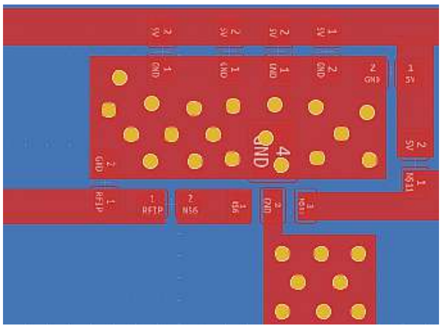


Fig. 3.9: Circuit Schematic of Amplifier

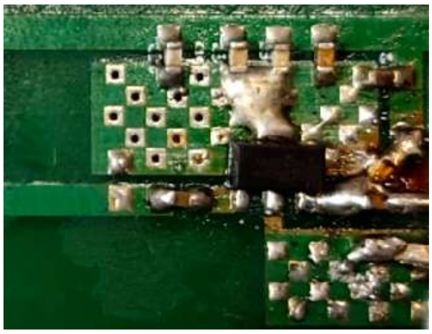
S-parameters for the same, given in Fig. 3.11, show that the amplifier performs very well throughout the operating band. The values of S11 and S22 are around –20 dB, indicating good input and output matching with minimal signal reflection throughout the band. The gain (S21) remains around 20 dB, showing consistent amplification across frequencies. PCB is designed for this circuit and the corresponding layout and final PCB are shown in Fig. 3.10(a) and 3.10(b) respectively.

3.2.3.2 Measured Results

The experimental setup consists of a Scientech 4077 - DC power supply and Rhode & Schwartz FPH.06 spectrum analyzer. The output of the VCO, shown in Fig. 3.8, is fed to the power amplifier circuit, and the resulting output spectrum seen in spectrum analyzer is given in Fig. 3.12. It can be observed that the output power level lies around 18 dBm, which confirms the amplifier's gain of approximately 20 dB.



(a) PCB layout for RF Amplifier



(b) Populated PCB of RF Amplifier

Fig. 3.10: PCB Layout and Fabricated PCB of the RF Amplifier

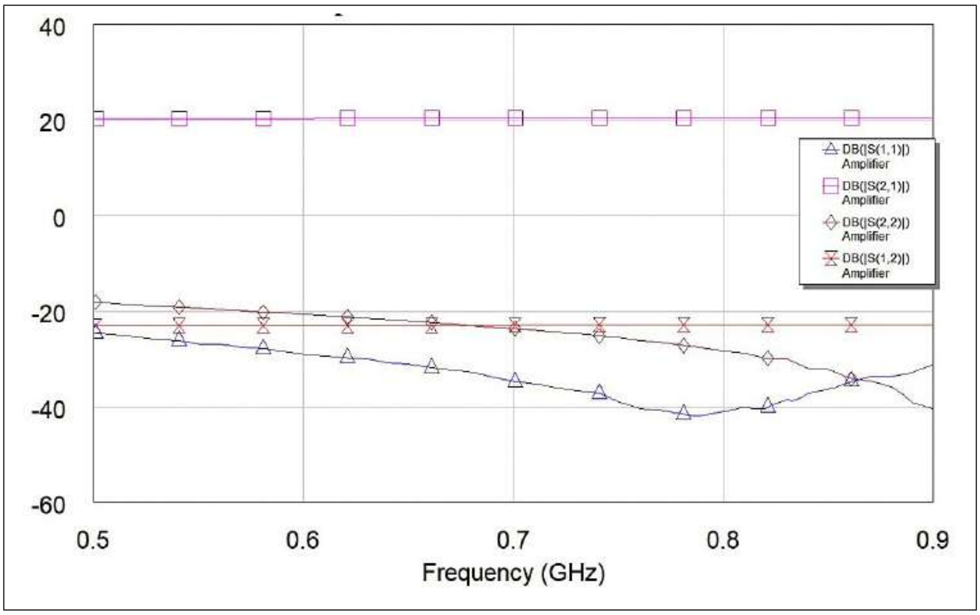


Fig. 3.11: S-parameters of Amplifier seen in AWR Microwave Office

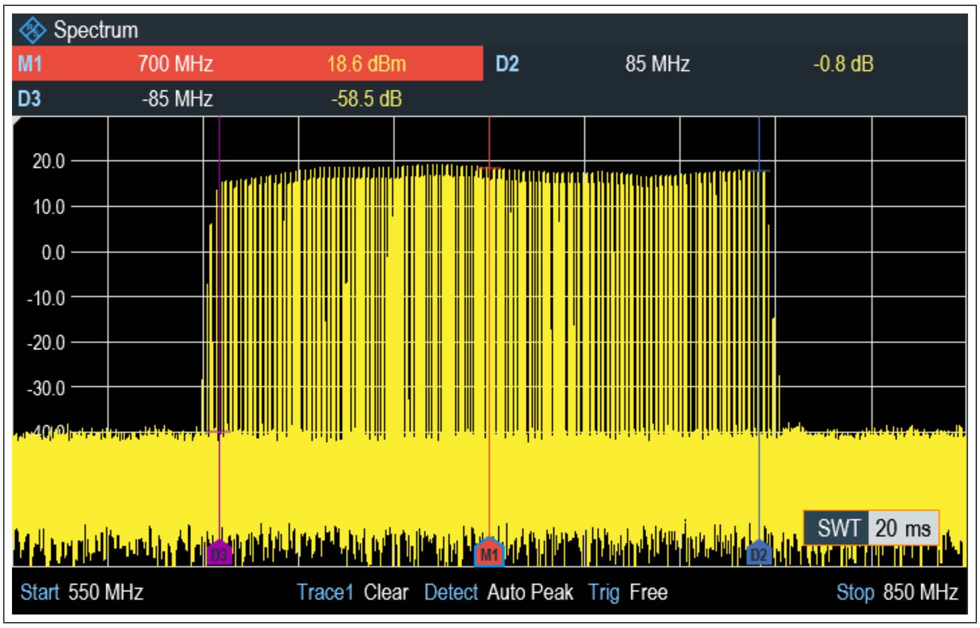


Fig. 3.12: Output spectrum of Amplifier fed with VCO signal (Output level: 18 dBm)

3.2.4 Power Divider

The power divider circuit is used to generate the reference local oscillator (LO) signal for the I-Q demodulator, which is responsible for demodulating the received signal to extract information about the target object. This module is placed between the amplifier and the antenna, and it functions by directing a portion of the transmitted signal toward the receiver.

3.2.4.1 Design Specifications

The ADL5380 I-Q demodulator IC [22], discussed in detail in later sections, requires an LO input power level between –6 dBm (0.25 mW) and +6 (4 mW) dBm, with a recommended value of approximately 0 dBm (1 mW) for optimal demodulation performance. Therefore, the power divider should be designed such that one of its output ports delivers approximately 0 dBm (1 mW) of power to serve as the LO input, while the input is driven by a power amplifier providing around 18 dBm (70 mW). The remaining power, which is largely unaffected, is directed to the other output port and to the antenna for transmission.

3.2.4.2 Implementation

For this purpose, an L-network lumped-element Wilkinson power divider [23], widely used for its simple and effective design, is employed. The circuit used for the same is shown in Fig. 3.13 [23]. The component values were optimized using the AWR Microwave Office software to achieve the best input and output matching and to obtain the desired output power levels of 0 dBm and 17 to 18 dBm in ports 3 and 2 respectively. The optimized S-parameters of the same are shown in Fig 3.14.

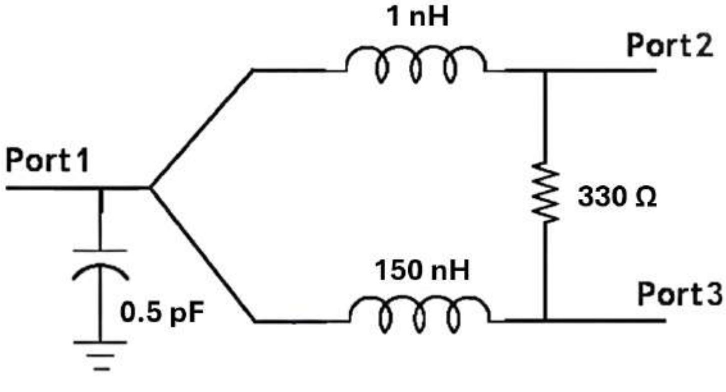


Fig. 3.13: Circuit schematic of Wilkinson divider used

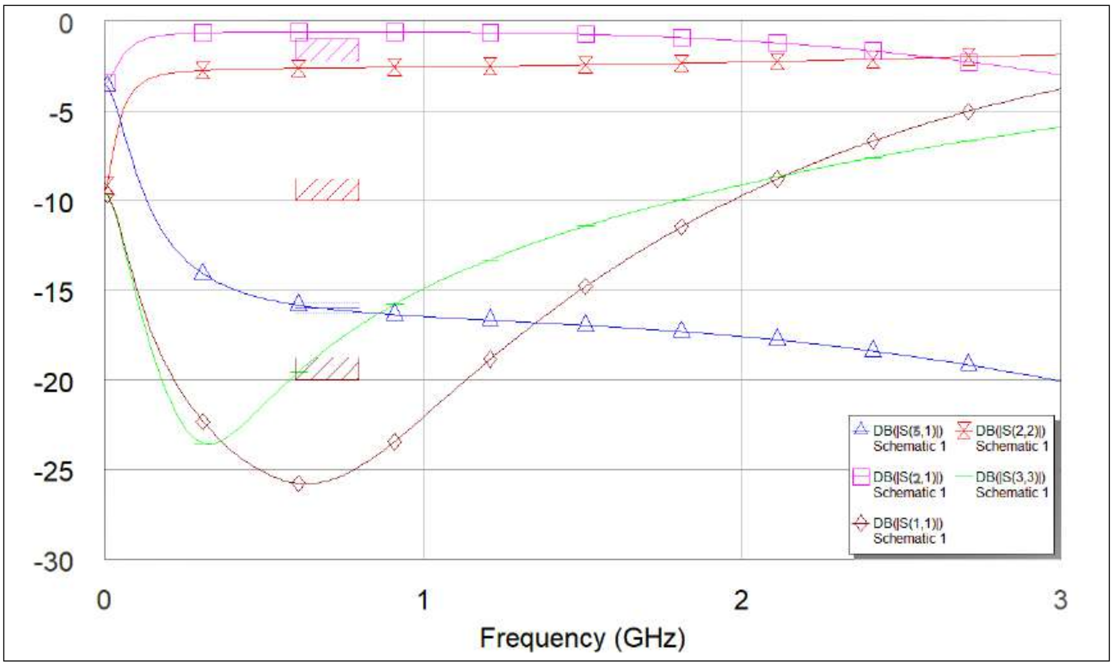
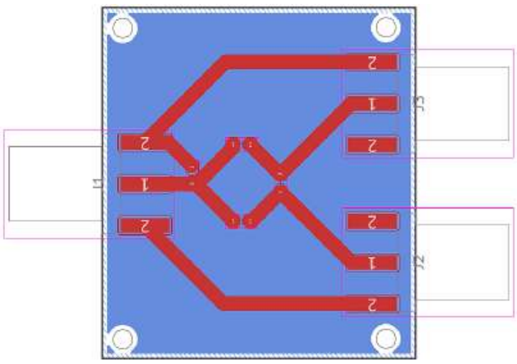
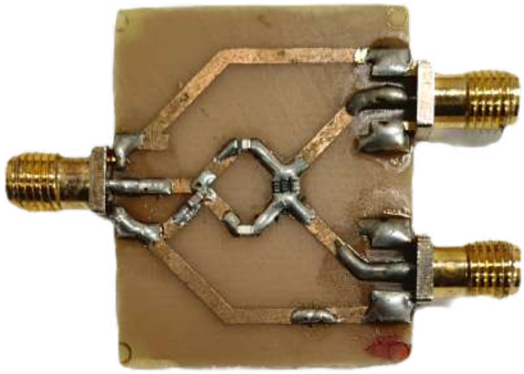


Fig. 3.14: Optimized S-parameters of power divider seen in AWR Microwave Office

PCB for the same is designed and fabricated locally. The PCB layout and the final populated PCB are shown in Fig. 3.15(a) and Fig. 3.15(b) respectively.



(a) PCB layout for Power divider



(b) Populated PCB of Power divider

Fig. 3.15: PCB Layout and Populated PCB of the Power divider

3.2.4.3 Measured Results

The experimental setup consists of a Scientech 4077 DC power supply and Rhode & Schwartz FPH.06 spectrum analyzer and Inritsu S820E vector network analyzer (VNA). From the observed S-parameters of the power divider shown in Fig. 3.16, it can be seen that S31 is around - 18 dB in the band of operation as required. The S21 is around -0.7 dB as expected. S11 is less than - 18 dB in the entire band of operation, showing that the input is well matched.

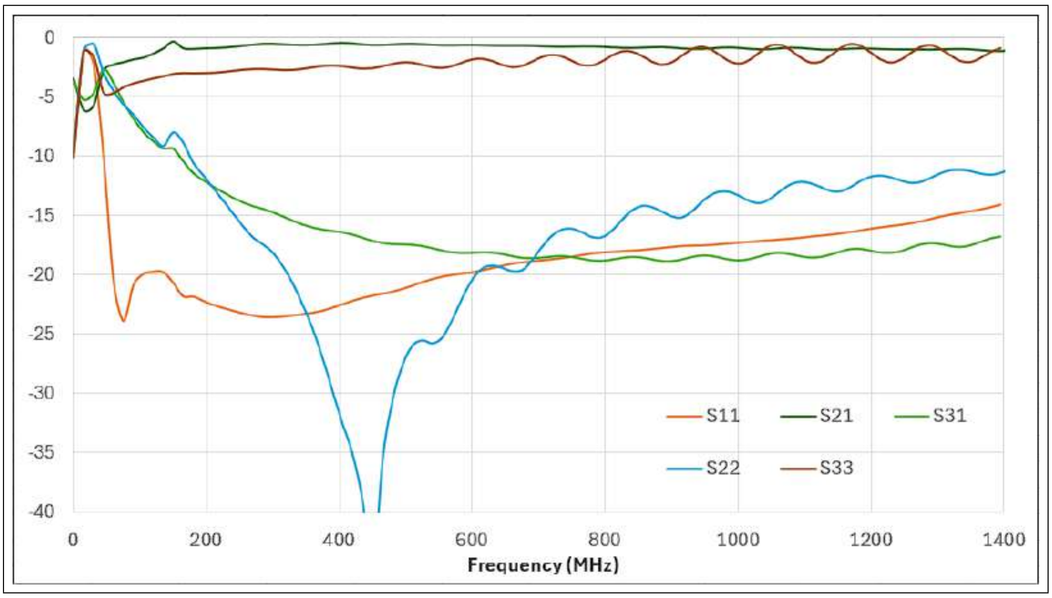


Fig. 3.16: S-parameters of power divider as seen in VNA

The power divider is driven by the output of the power amplifier, shown in Fig. 3.12. The output spectrum of port 3, measured using the spectrum analyzer, is presented in Fig. 3.17. As expected, the output power from port 3 is approximately 0 dBm, which is optimal for use as the LO signal. As the S21 observed is around -0.7 dB, the output power of port 2 is nearly equal to that of the input and is therefore good enough for transmission.

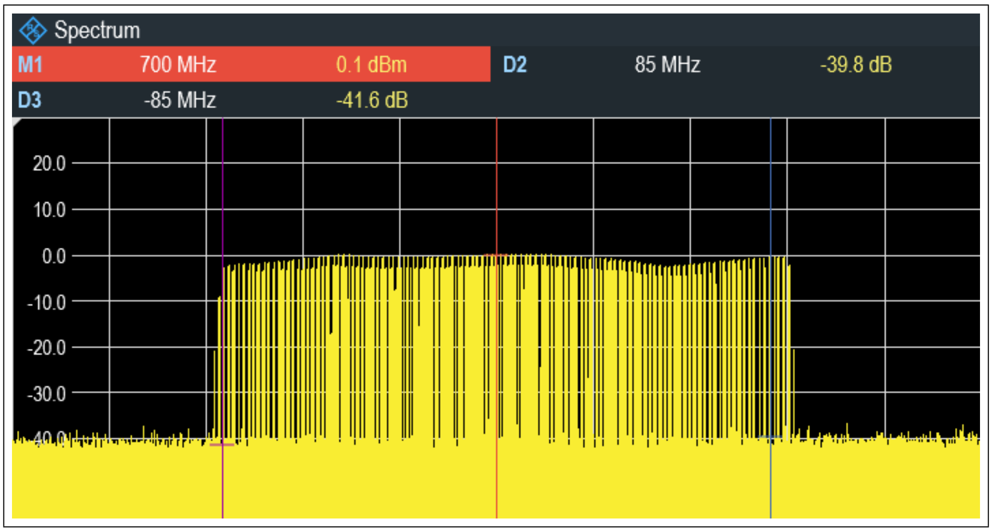


Fig. 3.17: LO signal output from Power Divider (Output level: 0 dBm)

3.3 Transmitting and Receiving Antennas

In Ground Penetrating Radar (GPR) systems, the choice of antennas used for transmission and reception plays a crucial role in determining the overall system performance, penetration capability, and signal fidelity. For this project, a **Log-Periodic Dipole Array (LPDA) antenna** shown in Fig. 3.18 has been used for both transmitting (Tx) and receiving (Rx) operations. This antenna was originally designed and fabricated by a senior student during his BTP project, and its wideband characteristics made it highly suitable for our radar experiments.

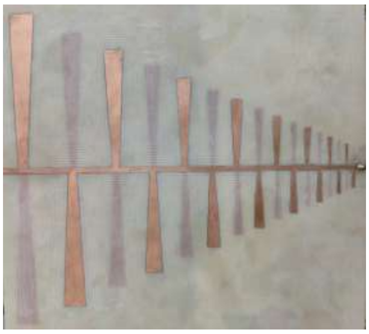


Fig. 3.18: Log-Periodic Dipole Array Antenna Used

3.3.1 Why LPDA Antenna?

GPR applications typically require antennas that maintain consistent radiation properties over a wide frequency band. The LPDA satisfies this requirement through its inherently frequency-independent behavior, achieved by its logarithmically scaled dipole elements.

- **Wideband Operation:** The antenna covers a frequency range from **450 MHz to 3.5 GHz**, making it capable of supporting both shallow and moderately deep subsurface profiling.
- **Directional Radiation Pattern:** The LPDA provides a focused beam, improving penetration into soil and reducing unwanted reflections and clutter from surrounding objects.
- **Single Antenna for Tx/Rx:** Using the same antenna type for both transmitting and receiving ensures symmetry in system response and reduces calibration complexity.
- **Stable Input Impedance:** Its geometry naturally provides a more uniform impedance profile across the operating band, allowing better matching with RF front-end hardware.

3.3.2 Working Principle

The LPDA is composed of multiple dipole elements of varying lengths and spacings, arranged such that only a subset of elements becomes active at any given frequency. When excited with an RF signal, the antenna behaves as if only the dipoles resonant to that specific frequency range are contributing to radiation. This creates a traveling-wave type current distribution, enabling stable gain and beam characteristics over a broad range.

The longest dipoles support radiation at the lower end of the spectrum (near 450 MHz), while the shorter dipoles are responsible for higher-frequency operation up to 3.5 GHz.

3.3.3 S-Parameter Measurements

To characterize the antenna performance, S-parameter measurements were carried out using an Inritsu S820E vector network analyzer (VNA). The reflection coefficient (S_{11}) provides insight into the impedance matching between the antenna and the feed line. A lower S_{11} value (typically below –10 dB across most of the band) indicates efficient power transfer and minimal reflection back into the transmitter. The measured S-parameters of the LPDA antennas used in this work is shown in Fig. 3.19. The measurement results confirm that the antenna maintains acceptable return loss across the intended 450 MHz–3.5 GHz band, validating its suitability for our GPR application.

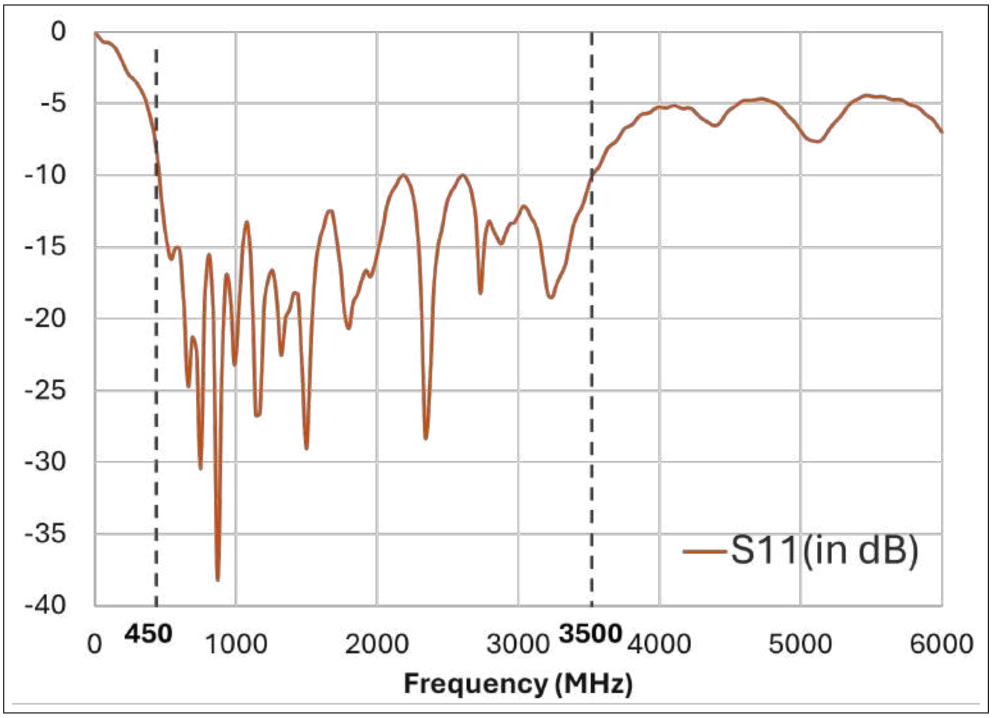


Fig. 3.19: Measured S-parameters of the LPDA antenna used in the GPR system.

3.4 Receiver Chain Design and Implementation

The design, implementation, and test results of various RF modules involved in the receiver chain are discussed in this section. These modules include the Low Noise Amplifier (LNA), Bandpass Filter (BPF), I-Q Demodulator and signal processing circuitry. All active modules are designed to operate with a 5 V supply, minimizing the need for additional voltage conversion circuitry.

3.4.1 Low Noise Amplifier (LNA)

LNA provides low noise figure, high gain and stable operation throughout the frequency range of operation. It plays a crucial role in receiver design by amplifying very weak input signals (often around –30 dBm or lower) without significantly increasing noise, thereby maintaining the desired signal-to-noise ratio (SNR). It directly influences receiver's sensitivity and decides the receiver's dynamic range, defined by the difference between the weakest and strongest signals it can handle.

3.4.1.1 Design and Implementation

For the realization of LNA, MAAL-011207-QR3000 IC [24] was chosen for its high gain of 25 dB with Output 1 dB Compression Point (P1dB) of 18.7 dBm and a low noise figure of 0.6 dB. Considering future applications, the wideband configuration of this IC recommended in the datasheet [24], covering 0.15 to 2.2 GHz, is adopted. The schematic of the implemented circuit is shown in Fig. 3.20. The PCB layout and populated PCB of the same are shown in Fig. 3.21(a) and Fig. 3.21(b) respectively. Multiple vias were used in PCB to provide improved grounding between the top and bottom copper planes in the CPWG (Coplanar Waveguide with Ground) structure.

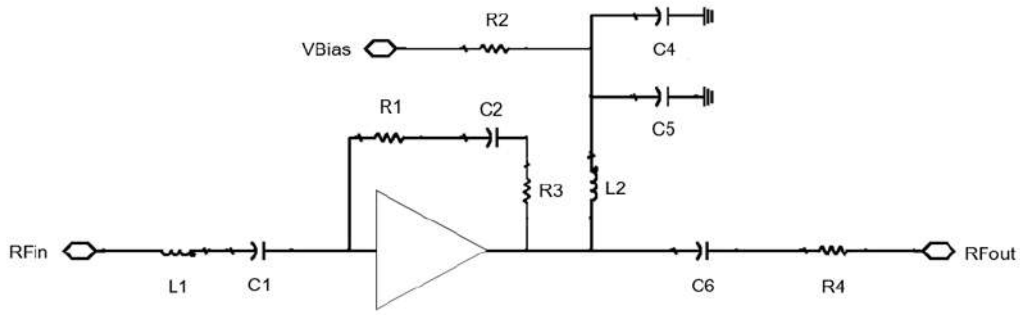
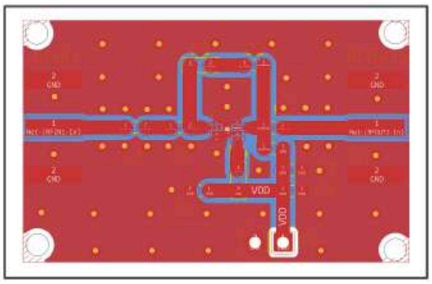
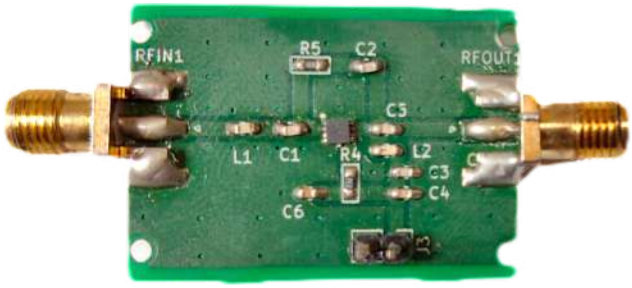


Fig. 3.20: Circuit schematic of MAAL-011207 LNA



(a) PCB layout for MAAL-011207 LNA



(b) Populated PCB of MAAL-011207 LNA

Fig. 3.21: PCB Layout and Populated PCB of the MAAL-011207 LNA

3.4.1.2 Measured Results

The experimental setup consists of a Scientech 4077 DC power supply, Rhode & Schwartz FPH.06 spectrum analyzer, Inritsu S820E vector network analyzer (VNA) and Ettus USRP B210 Software defined radio (SDR). The S-parameters measured using the VNA, shown in Fig. 3.22, confirm a gain of approximately 25 dB across the designed frequency range of 610 to 790 MHz. The observed S11 is below -10 dB, indicating excellent input impedance matching and S21 remains greater than 18 dB through 0.15 to 2.2 GHz band demonstrating the desired amplification performance.

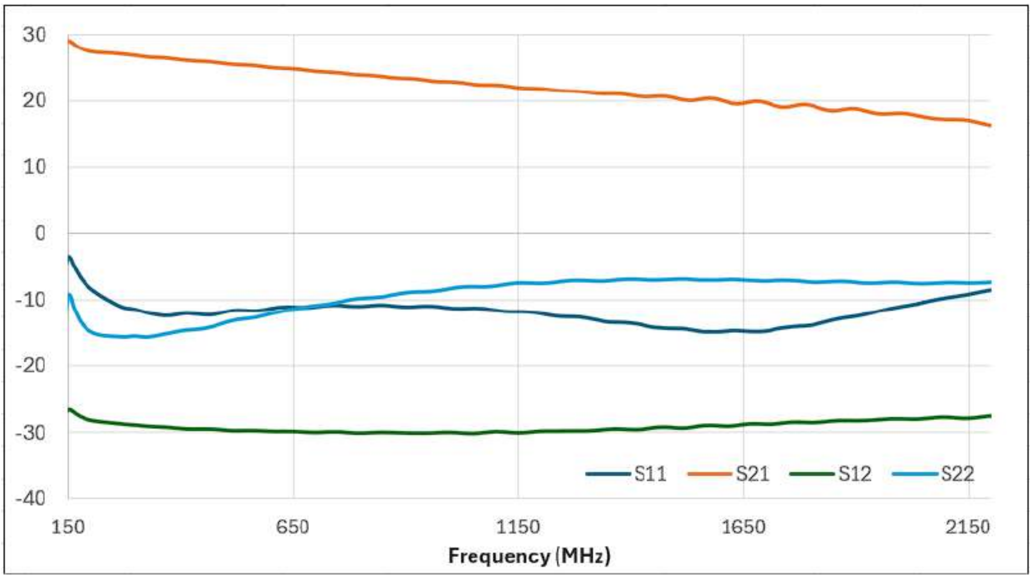


Fig. 3.22: S-parameters of LNA module as seen in VNA

A 700 MHz signal with a power level of - 9.3 dBm, as shown in Fig. 3.23, was given as an input to the LNA from the SDR. The output power for this input, observed in spectrum analyzer is around 15 dBm and is shown in Fig. 3.24. The measured gain of the LNA is approximately 24.3 dB, which is within 1 dB of the expected value and thus considered well within acceptable limits. To further validate its performance, the amplifier was tested with input signals at 600 MHz and 800 MHz. In both cases, the gain remained consistent, varying by less than 1 dB across the entire bandwidth of interest. This demonstrates the amplifier's stable operation, good linearity, and reliable performance across the target frequency range.

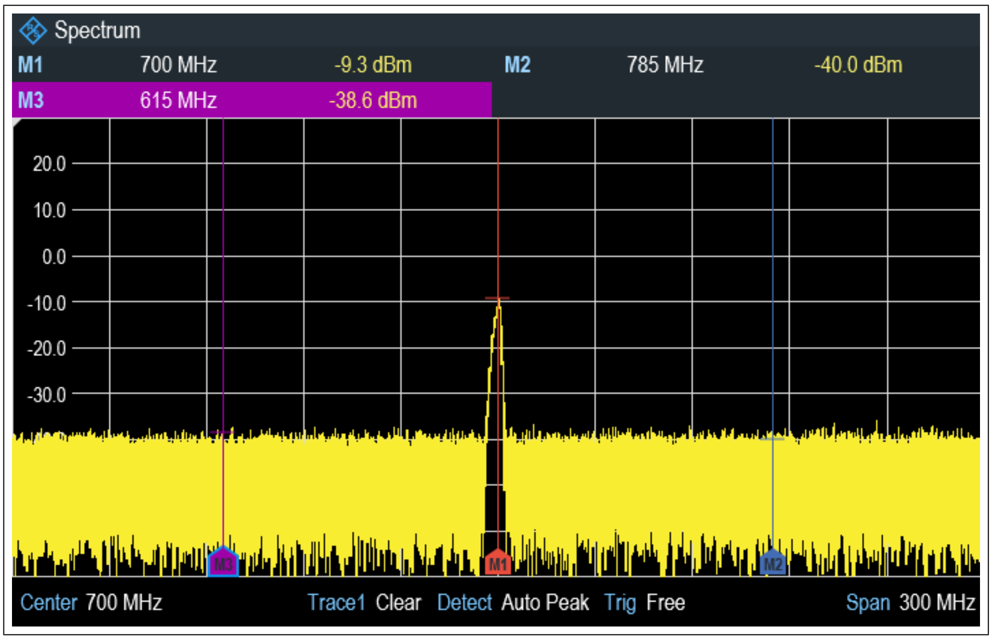


Fig. 3.23: Input of LNA seen in spectrum analyzer (Input level: -9.3 dBm, 700 MHz)

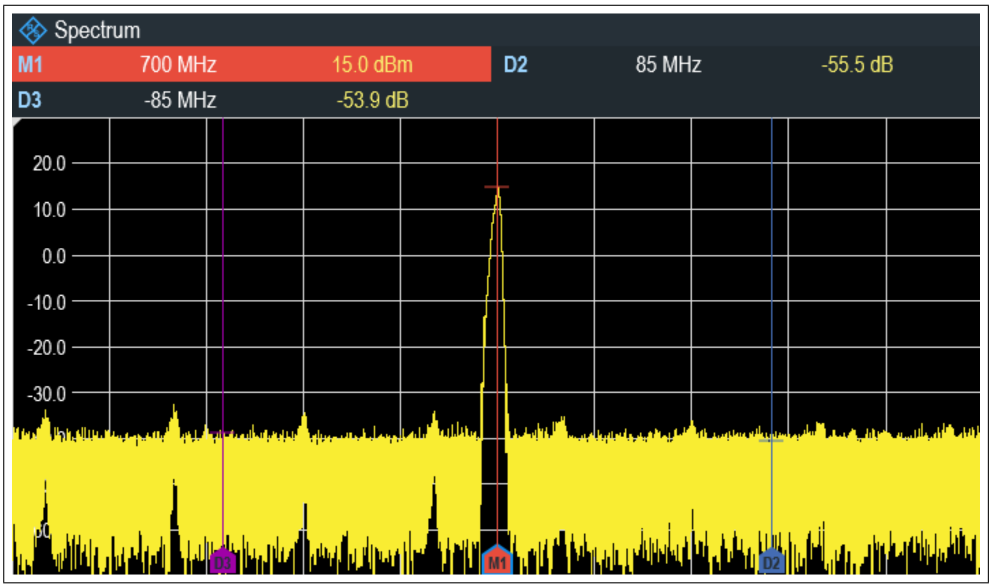


Fig. 3.24: Output of LNA seen in spectrum analyzer (Output level: 15 dBm, 700 MHz)

3.4.2 Band-pass Filter (BPF)

Since LNA was designed to amplify all frequencies from 0.15 to 2.2 GHz, it also amplifies frequencies in other bands that are picked up by the receiving antenna. This may cause problems in demodulation and detection of the target surface. Bandpass Filter (BPF) is mainly used to prevent this by allowing only signals in our band of interest to pass through while attenuating the remaining signals. The pass band refers to the band of frequencies that the BPF allows without significant attenuation, and the stop band refers to the band of frequencies that the BPF attenuates significantly.

3.4.2.1 Design Specifications

The main design parameters for BPF are the pass-band gain flatness and the transition band roll-off. The final BPF design is required to exhibit a low insertion loss of less than 1 dB within the pass-band, while maintaining a high attenuation greater than 20 dB in the stop-bands. In addition, both input and output ports should be properly matched to ensure minimal return loss and efficient power transfer.

3.4.2.2 Design and Implementation

Generally, BPFs operating at microwave frequencies are implemented using microstrip line structures. However, at a center frequency of 700 MHz, the physical length of the required microstrip lines becomes quite large. Moreover, when using an FR-4 substrate, the dielectric and conductor losses significantly increase, leading to higher insertion loss. Therefore, a lumped element coupled resonator band-pass filter with a Chebyshev response is designed for operation around 700 MHz. The schematic of the filter is shown in Fig. 3.25. The filter consists of alternating series capacitors and shunt LC resonators forming a seven-pole structure, providing sharper attenuation outside the pass-band compared to a Butterworth filter.

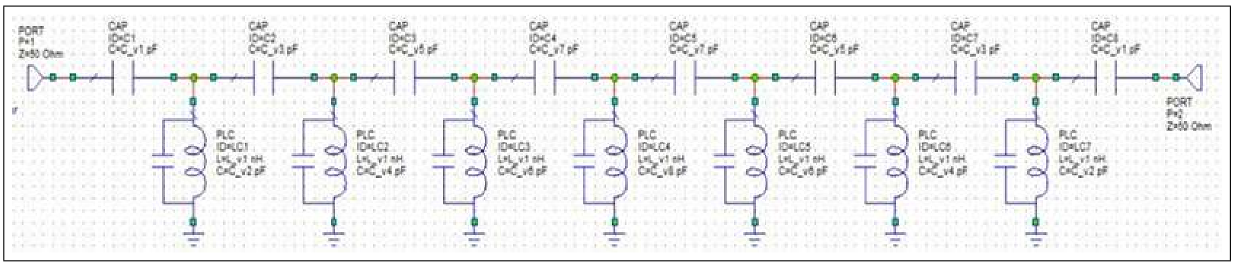


Fig. 3.25: Schematic Circuit of Bandpass Filter

The values of the L and C elements are tuned and optimized in AWR Microwave Office to get the optimal S-parameters as shown in Fig. 3.26. From Fig. 3.26, it can be observed that the 3 dB pass-band of the filter extends approximately from 620 MHz to 770 MHz, resulting in a total bandwidth of about 150 MHz, which is slightly narrower than the 180 MHz bandwidth of the transmitter. This reduction in bandwidth was an intentional design trade-off to minimize pass-band ripple and improve overall flatness. Nevertheless, a 150 MHz bandwidth remains adequate to provide good resolution for subsurface target detection.

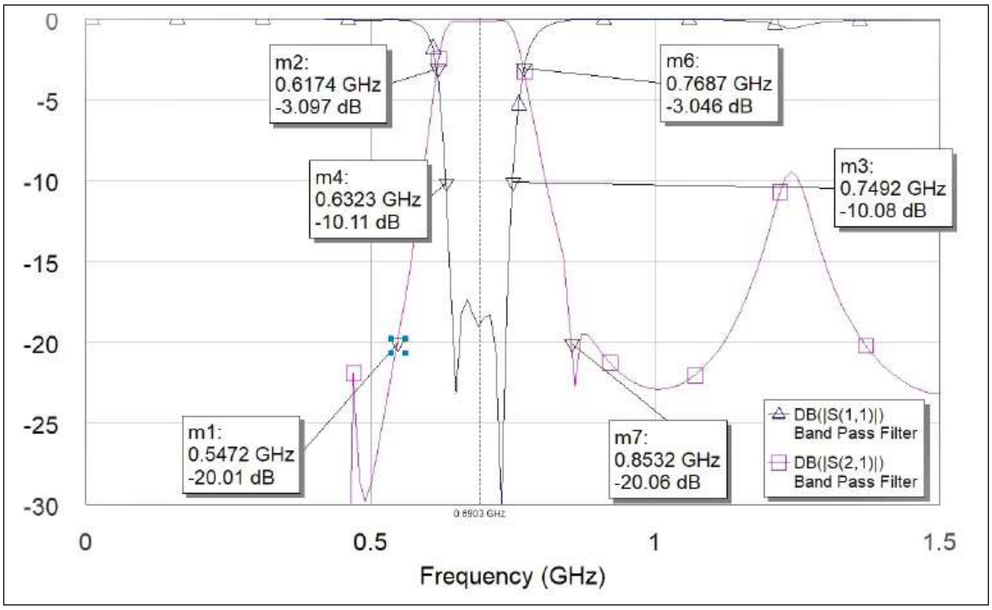


Fig. 3.26: Simulated S-parameters of BPF in AWR Microwave Office

PCB layout for the BPF circuit shown in Fig. 3.25, designed in KiCad, is given in Fig. 3.27. Populated PCB of the same is shown in Fig. 3.28. Multiple vias were used to provide improved grounding between the top and bottom copper planes in the CPWG structure.

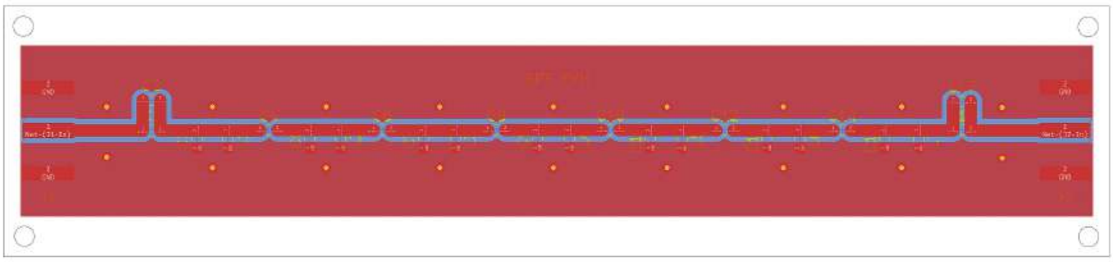


Fig. 3.27: PCB layout of Bandpass filter

Fig. 3.28: Populated PCB of Bandpass filter

3.4.2.3 Measured Results

The experimental setup consists of a Inritsu S820E vector network analyzer (VNA). The measured S-parameters of the band-pass filter obtained from the VNA are shown in Fig. 3.29. From this figure, it can be observed that the pass-band insertion loss (S21) is around 2 to 3 dB. This loss is due to the dielectric losses in FR-4 substrate and the non-ideal behavior of L and C components that are not included in simulation. The stop-band attenuation achieved is over 20 dB. Furthermore, the input and output return loss (S11 and S22) are below -10 dB nearly throughout the pass-band signifying good impedance matching. In conclusion, the measured performance of the band-pass filter satisfies the requirements for the subsequent stage of the system and is good enough to proceed.

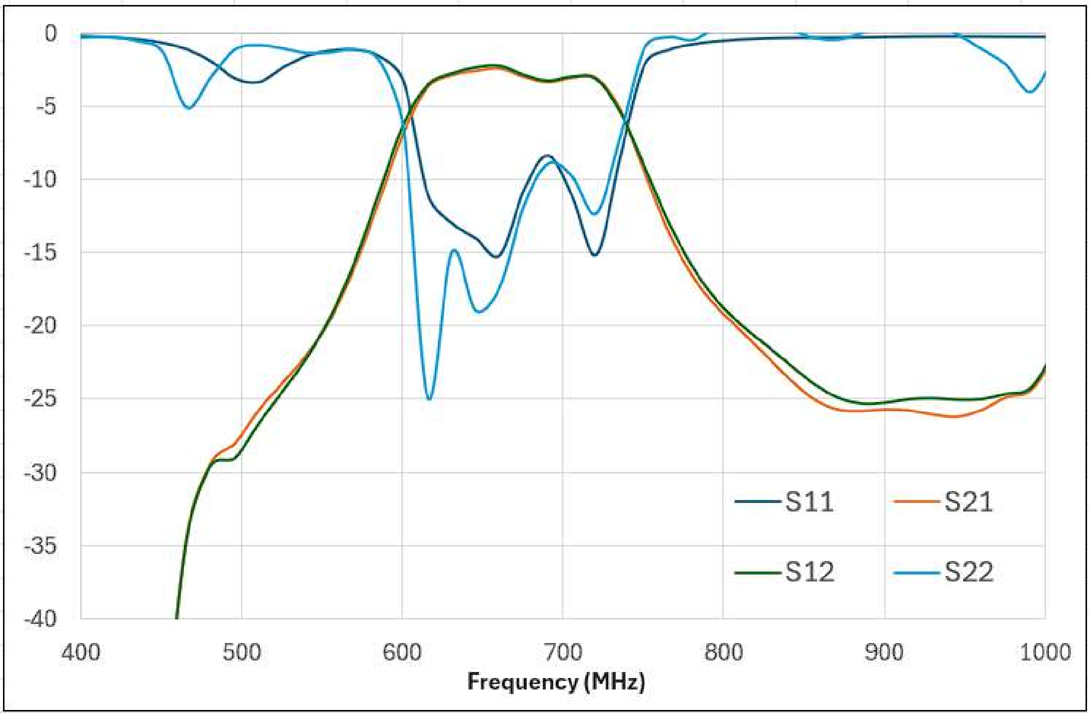


Fig. 3.29: S-parameters of BPF as seen in VNA

3.4.3 I-Q Demodulator

The received signal contains data on any dielectric discontinuities in its path from which we can detect the presence of landmines. Demodulation is performed to extract information about the variations in the received signal relative to the transmitted one, thereby identifying objects responsible for those changes. In the I-Q demodulator, this is achieved by mixing the portion of the transmitted signal with the received signal. The output is then filtered to get the baseband in-phase (I) and quadrature (Q) components, which differ from each other by a phase angle of 90°. The frequency of these I, Q components is equal to the difference in the frequencies of the reference LO and received RF signals. By analyzing the frequency and amplitude of the baseband I and Q signals, the nature of the discontinuity and its distance from the surface can be accurately determined. Due to its low cost and simple implementation requiring minimal external components, the ADL5380 I/Q Demodulator [22] from Analog Devices is chosen for the demodulator implementation.

3.4.3.1 Implementation with ADL5380

The ADL5380 is a broadband quadrature I-Q demodulator that covers a frequency range from 400 MHz to 6 GHz [22]. It is suitable for use in many communication receivers, performing quadrature demodulation directly to baseband frequencies. It accepts differential LO and RF inputs and downconverts them to differential baseband I and Q output signals. Block diagram for ADL5380 is shown in Fig. 3.30 [22].

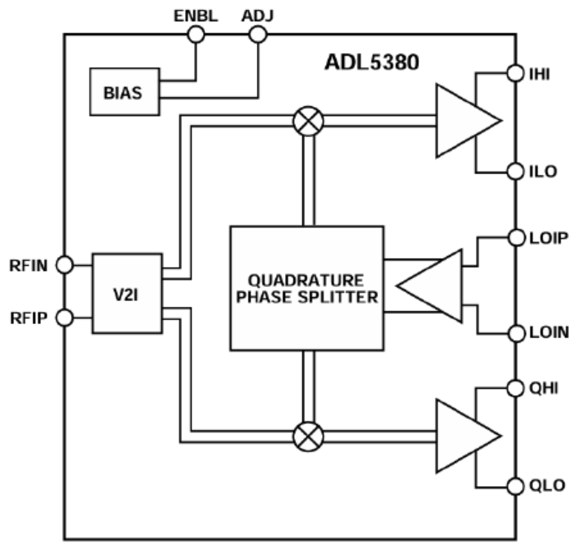


Fig. 3.30: Block diagram of I-Q demodulator IC (ADL5380)

The LO input of the demodulator is driven by the power divider discussed in section 3.2.4. As the recommended power level for LO signal is 0 dBm [22], power divider in section 3.2.4 is designed to output a power level of around 0 dBm from its output. The single-ended outputs from the power divider and band-pass filter, designed in the previous sections, need to be converted into differential signals before interfacing with the ADL5380. For this purpose, the Mini-Circuits RF balun TCM1-63AX+, covering wideband applications from 400 MHz to 6 GHz [25], is employed at both the LO and RF input ports of the demodulator as shown in Fig. 3.31 [22]. The application circuit provides flexibility to obtain the baseband output either in differential form directly or as single-ended signals through an RF balun, as illustrated in Fig. 3.31. The TCM9:1+ RF balun [26] was chosen since it provides a recommended total impedance of 500 Ω for baseband outputs when connected to a 50 Ω output load.

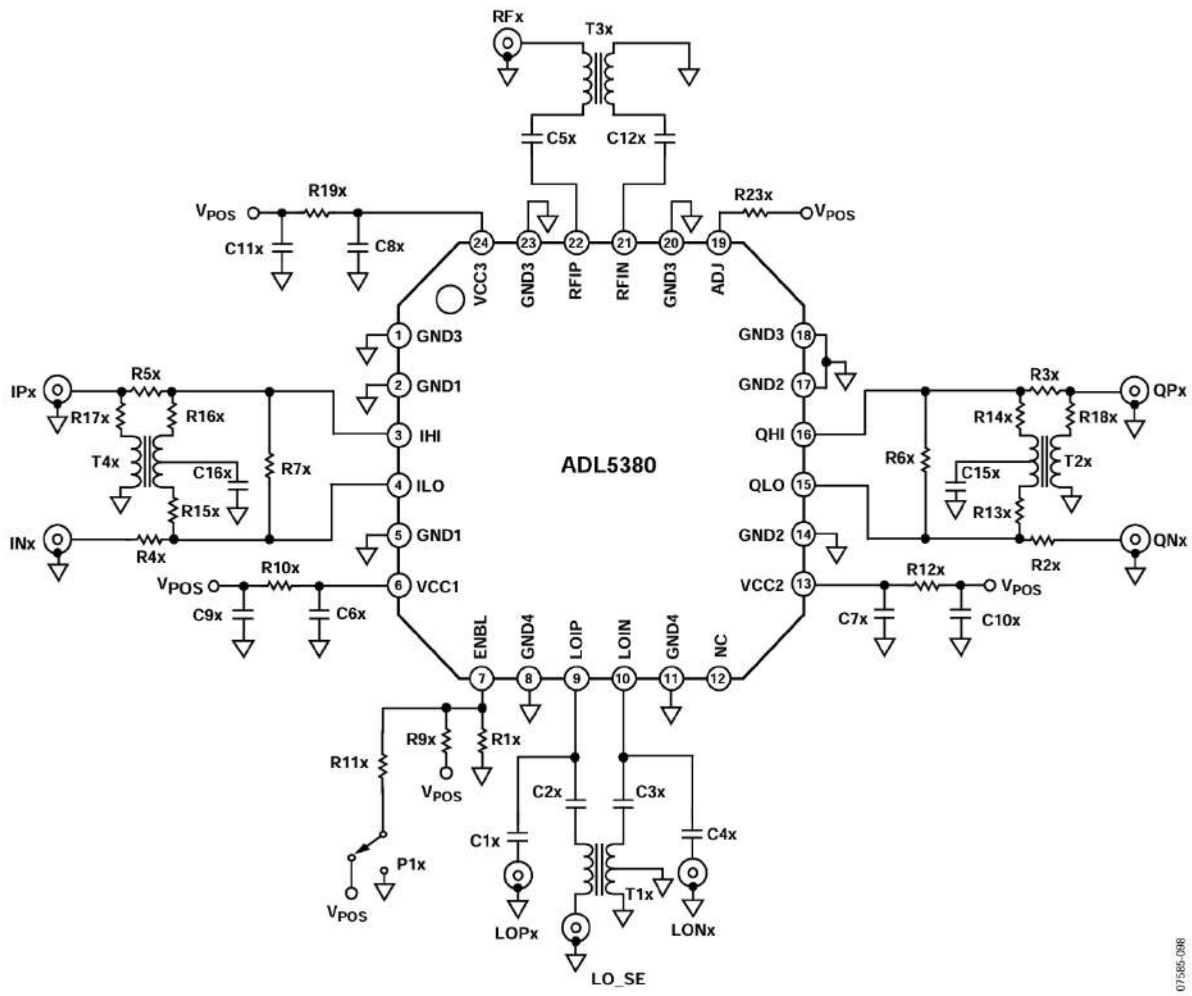
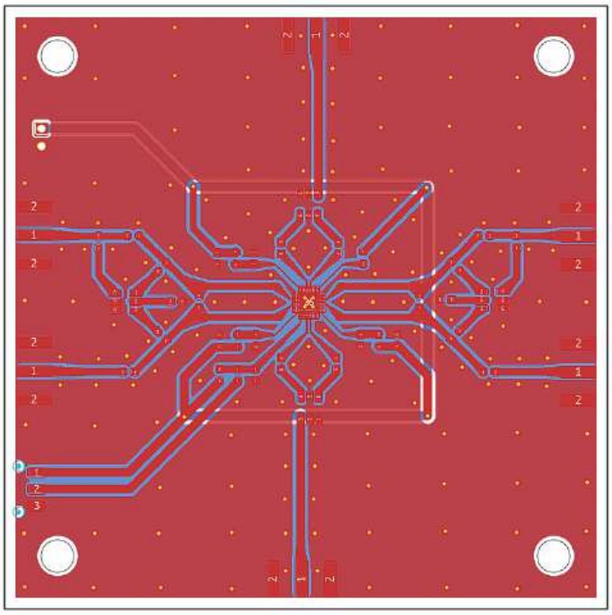
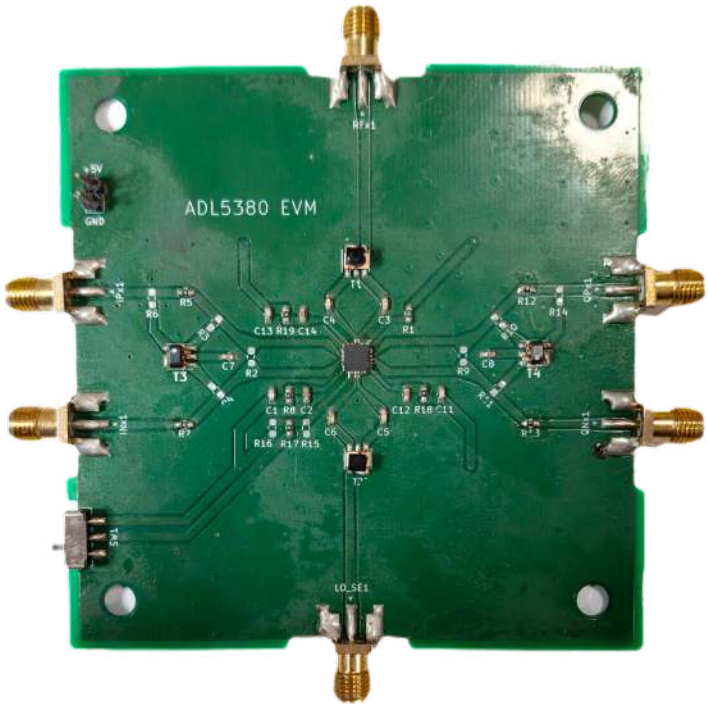


Fig. 3.31: Application circuit schematic of I-Q demodulator IC (ADL5380)

The ADJ and ENBL pins, which are directly connected to the internal bias circuitry, are configured according to the recommendations provided in the datasheet [22]. Multiple decoupling capacitors are connected to the VCC pins to effectively suppress any supply noise propagation. The corresponding PCB for the application circuit in Fig. 3.31 was designed in KiCad and the layout for the same is given in Fig. 3.32(a). Multiple vias were used to provide improved grounding between the top and bottom copper planes in the CPWG structure. The fabricated PCB is shown in Fig. 3.32(b).



(a) PCB layout for I-Q demodulator (ADL5380)



(b) Populated PCB of I-Q demodulator (ADL5380)

Fig. 3.32: PCB Layout and Populated PCB of the I-Q Demodulator (ADL5380)

3.4.4 Signal Conditioning and Processing

The received signal from the antenna contains weak reflections from the buried objects along with clutter and noise. These signals should be properly conditioned before digitization to enhance their quality. This is achieved by filtering the I and Q baseband outputs of the demodulator and then digitizing them using ADCs.

3.4.4.1 Required LPF and ADC specifications

The cut-off frequency of Low-pass Filer (LPF) used and the sampling rate of ADC for optimal digitization are highly dependent on the beat frequency present in the I and Q components. The beat frequency for an up-sweep FMCW triangular chirp is approximated as

$$f_b = \frac{2RB\sqrt{\varepsilon_r}}{c\,T_{\text{sweep,up}}}, \qquad T_{\text{sweep,up}} = \frac{1}{2f_{\text{chirp}}},$$

where B is sweep bandwidth, R is range, ε_r is soil relative permittivity, c is light speed and f_{chirp} is the chirp frequency.

For a given beat frequency, the cut-off frequency of the LPF should be slightly higher, typically 1.1 to 1.2 times the beat frequency, to ensure that the desired signal passes through without distortion. According to the Nyquist criterion, the ADC sampling rate must be at least twice the beat frequency to accurately capture the signal without aliasing. For a maximum range of 60 cm, sweep bandwidth of around 180 MHz and chirp frequency of 9.75 kHz from TWG, various beat frequency values corresponding to different soil types, along with the required LPF cutoff frequencies, are summarized in Table 3.1.

Soil ε_r	f_b **(kHz)**	$f_c \approx 1.1 f_b$ **(kHz)**
1 (air)	14.040	15.444
4 (dry soil)	28.080	30.888
9 (moist soil)	42.120	46.332
25 (very wet)	70.200	77.220

Table 3.1: Beat frequency and corresponding LPF cutoff frequency for different soil types

Considering the worst-case scenario, a fourth-order low-pass filter is designed with a cutoff frequency of 75 kHz. Accordingly, the minimum sampling rate required for the ADC should be at least twice the maximum beat frequency, i.e., 140 kSPS, to satisfy the Nyquist criterion.

3.4.4.2 Selection of appropriate Micro-controller

From the discussion in the previous section, the minimum sampling rate required for the ADCs is determined to be 140 kSPS. This sampling rate isn't very demanding and can easily be handled by the inbuilt ADCs of a DSP-based microcontroller. In this work, the STM32G431KB is used for both sampling and signal processing. It offers Dual-12-bit ADCs capable of sampling up to 4 MSPS, along with an ARM Cortex-M4 core that supports floating-point operations and DSP instructions. These features make it powerful enough to perform real-time digital filtering and FFT computations efficiently. Its compact size, low power consumption, and strong processing performance make it a perfect fit for FMCW signal demodulation in this GPR setup.

3.4.4.4 Data Acquisition

The overall data acquisition flow is shown in Figure 3.33. The process begins with configuring the ADCs in differential mode and triggering conversions using an internal timer. DMA continuously stores the incoming samples into memory buffers so that no samples are missed at higher sampling rates. Once a buffer is filled, an interrupt is generated, and the microcontroller computes the instantaneous amplitude and phase using the stored I and Q samples. These values are then transmitted over UART to an external system. On the host side, the received data is logged and visualized using a Python script. The script reads the amplitude and phase values in real time, updates plots for monitoring, and provides an option to store a reference trace for background subtraction. A complete version of both the STM32 firmware and the Python processing code is included in the appendix for reproducibility.

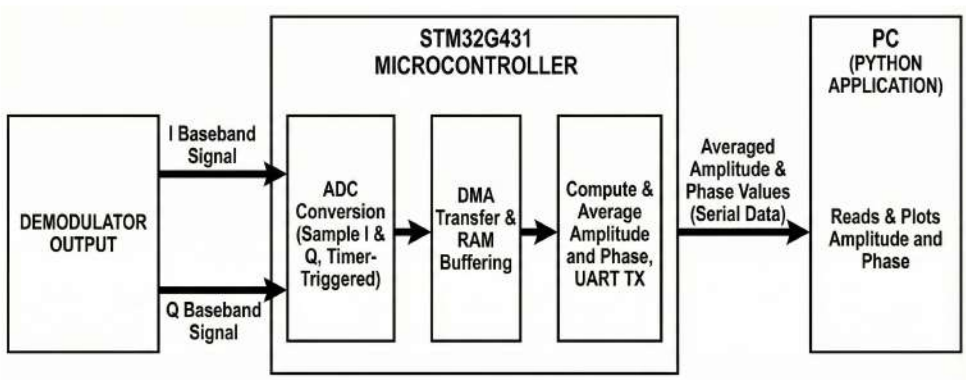


Fig. 3.33: Block diagram illustrating the data acquisition and processing flow

Chapter 4

Integration of GPR

4.1 Integration of Various Modules

The complete GPR system brings together the transmitter, receiver, and digital processing blocks on a compact hardware platform. Each module plays a key role in generating, capturing, and interpreting the FMCW radar echoes. The overall signal flow is briefly outlined below, followed by a photograph of the assembled setup in Fig. 4.1.

Transmitter

The transmitter chain starts with generating a stable Frequency Modulated Continuous Wave (FMCW) signal. A Voltage-Controlled Oscillator (VCO) serves as the main RF source, while a Triangular Wave Generator (TWG) sets the instantaneous frequency, creating the up–down sweep across the operating band. In systems requiring more flexibility, this triangular sweep can also be produced digitally using a microcontroller and a DAC.

To achieve the required penetration depth, the VCO output is boosted using an RF power amplifier. The amplified signal is then routed through a power divider: one output drives the transmitting antenna, and the other provides the Local Oscillator (LO) signal to the IQ demodulator in the receiver chain. This arrangement keeps the transmitted and received signals phase-coherent, which is crucial for accurate beat-frequency extraction.

Receiver

The receiver chain starts with the antenna capturing the weak echoes reflected from subsurface targets. These signals are first boosted using a Low Noise Amplifier (LNA) to improve the overall signal-to-noise ratio. A bandpass filter placed after the LNA removes out-of-band interference and reduces the effect of environmental noise.

The filtered signal is then mixed with the LO signal inside an IQ demodulator, which produces the in-phase (I) and quadrature (Q) baseband outputs. These baseband signals carry the beat frequency component from which target range information is obtained. The I and Q outputs are passed through low-pass filters to suppress high-frequency components and are subsequently digitized using dual ADC channels. A microcontroller processes the digitized data to compute amplitude and phase, enabling the identification of changes in the subsurface profile.

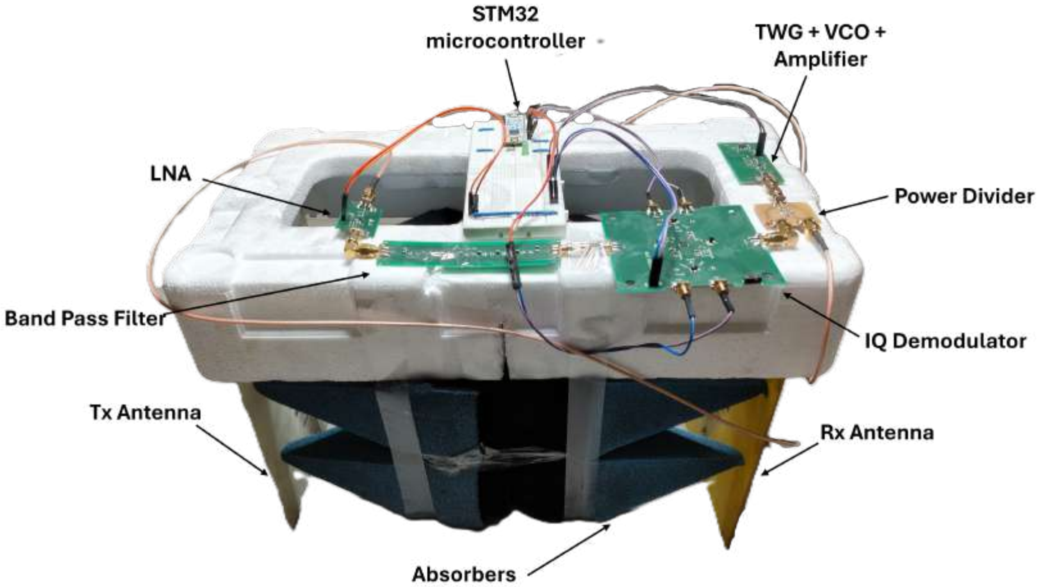


Fig. 4.1: Integrated FMCW-GPR hardware for Testing

4.2 Antenna Coupling Analysis Under Different Conditions

During the integration of the LPDA antennas into the GPR system, it was observed that the surrounding electromagnetic conditions influence the coupling between the antennas. To study this effect, S-parameter measurements were performed under three setups: **open air**, **with a metal plate placed between the antennas**, and with **RF absorbers** inserted between them. These configurations allow a direct comparison of coupling behavior under different levels of isolation. Coupling is evaluated using the transmission coefficient, which shows how much transmitted energy reaches the receiving antenna. High coupling can mask weak subsurface echoes, so assessing it is important for accurate radar operation.

4.2.1 Coupling in Open-Air Environment

In an open-air scenario, the antennas experience minimal obstruction and maximum free-space radiation, which naturally leads to higher mutual coupling. The measured S21 values showed noticeable pickup of the transmitted energy by the receiving antenna, with coupling levels around –20 dB, especially in the mid-band region where the antenna gain is maximum. This condition represents the worst-case coupling scenario and helps establish the baseline performance of the antennas without any environmental suppression.

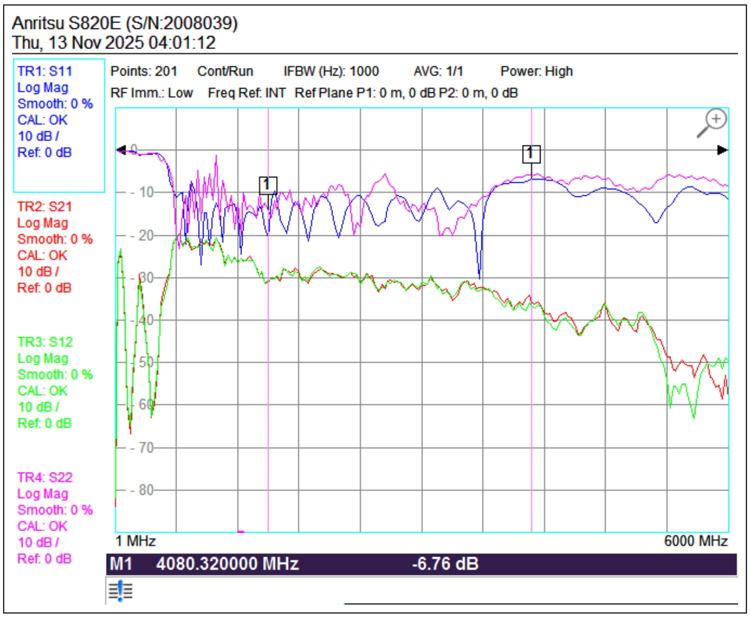


Fig. 4.2: Coupling Characteristics in Open Air

4.2.2 Coupling With a Metal Plate Inserted Between the Antennas

Placing a metal plate between the transmitting and receiving antennas noticeably changed their coupling behavior. Since the plate blocks and reflects most of the electromagnetic energy, it prevents direct propagation between the antennas. As a result, the measured S21 values decreased across most of the operating band, reaching approximately –30 dB, indicating significantly reduced coupling, as shown in Fig. 4.3. Small variations were still visible in both S11 and S21, mainly due to secondary reflections from the edges of the plate.

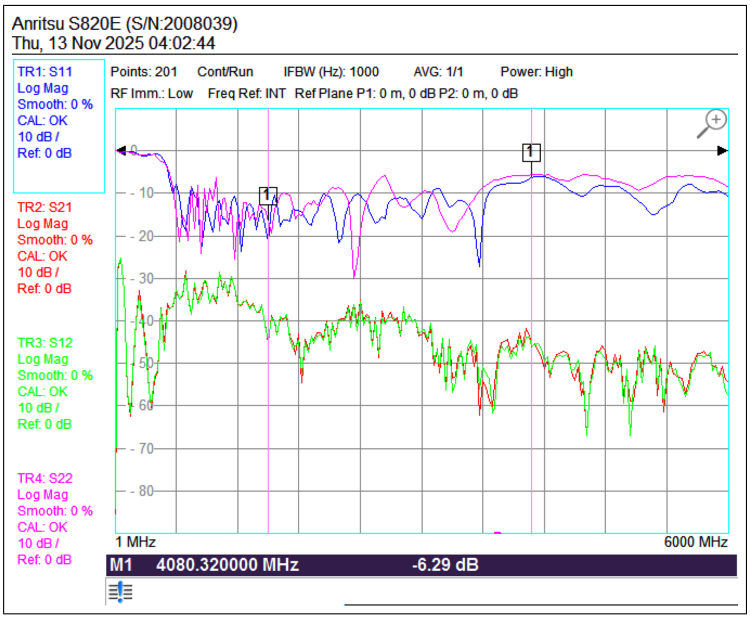


Fig. 4.3: Coupling with Metal Plate between Antennas

4.2.3 Coupling Reduction Using RF Absorbers

To reduce unwanted coupling, RF absorbers were inserted between the antennas. The absorbers suppress surface reflections and limit direct radiation from reaching the receiving antenna. As shown in Fig. 4.4, the S-parameter measurements exhibit a clear reduction in S21 across the band, reaching approximately –50 dB, confirming effective coupling suppression. The S11 response also becomes smoother due to minimized reflections. Overall, the absorbers improve isolation between the transmit and receive antennas, helping preserve weak target echoes.

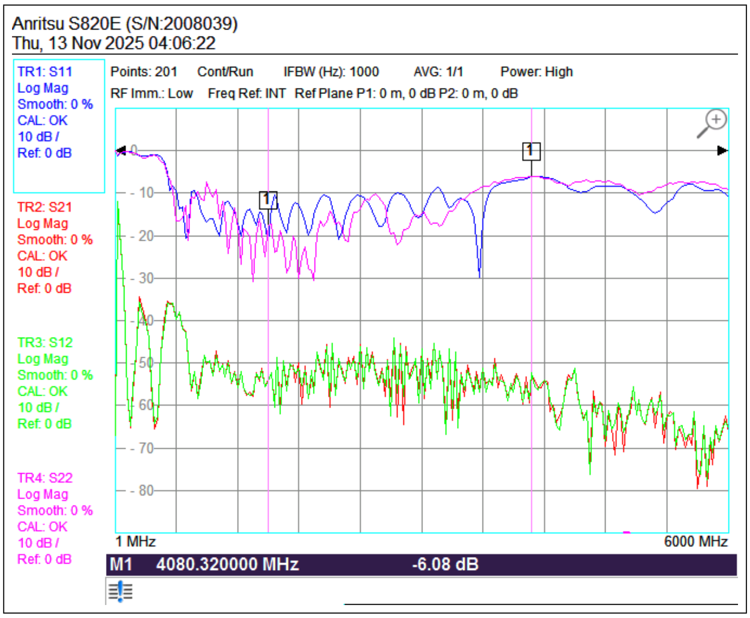


Fig. 4.4: Coupling Characteristics with RF Absorbers between Antennas

Chapter 5

Testing of Developed GPR Modules

This chapter presents the experimental evaluation of the FMCW-based GPR system developed in this work. The aim of the testing phase is to verify the functionality of the transmitter–receiver chain, assess the quality of the processed beat-frequency signals and demonstrate the ability of the system to detect different targets under controlled conditions. The experiments were carried out both in air and in soil to understand how target size, target material and surrounding medium influence the measured amplitude and phase response.

5.1 Experimental Setup

The complete test arrangement used for evaluating the system is shown in Fig. 5.1. The transmitter and receiver antennas were placed at a fixed separation, and the developed PCBS carrying the VCO, RF amplification stage, IQ demodulator and microcontroller was positioned close to the antennas to minimize cable losses. A Python-based data logging interface running on a laptop captured the amplitude and phase outputs in real time through a high-speed UART link.

A set of laboratory instruments was used throughout the measurements to monitor the signal chain and verify the performance of individual hardware blocks. The UTP33052 DC power supply provided regulated power to the RF and digital sections of the board. A Rhode & Schwartz FPH.06 spectrum analyzer was used to inspect the VCO output and observe the transmitted FMCW sweep. A Tektronix TDS1012C digital storage oscilloscope was employed for monitoring the triangular modulation waveform, LO signal and baseband I/Q outputs. Additionally, an Anritsu S820E vector network analyzer was used for antenna characterization and frequency response measurements. These instruments helped ensure that each subsystem was operating correctly before conducting detection experiments.

To maintain repeatability in measurements, all targets were placed along the antenna boresight at predefined positions. For soil experiments, a plastic container filled with soil was used, and targets were buried at known depths. Consistency in antenna height, soil compaction and alignment was maintained across all trials.

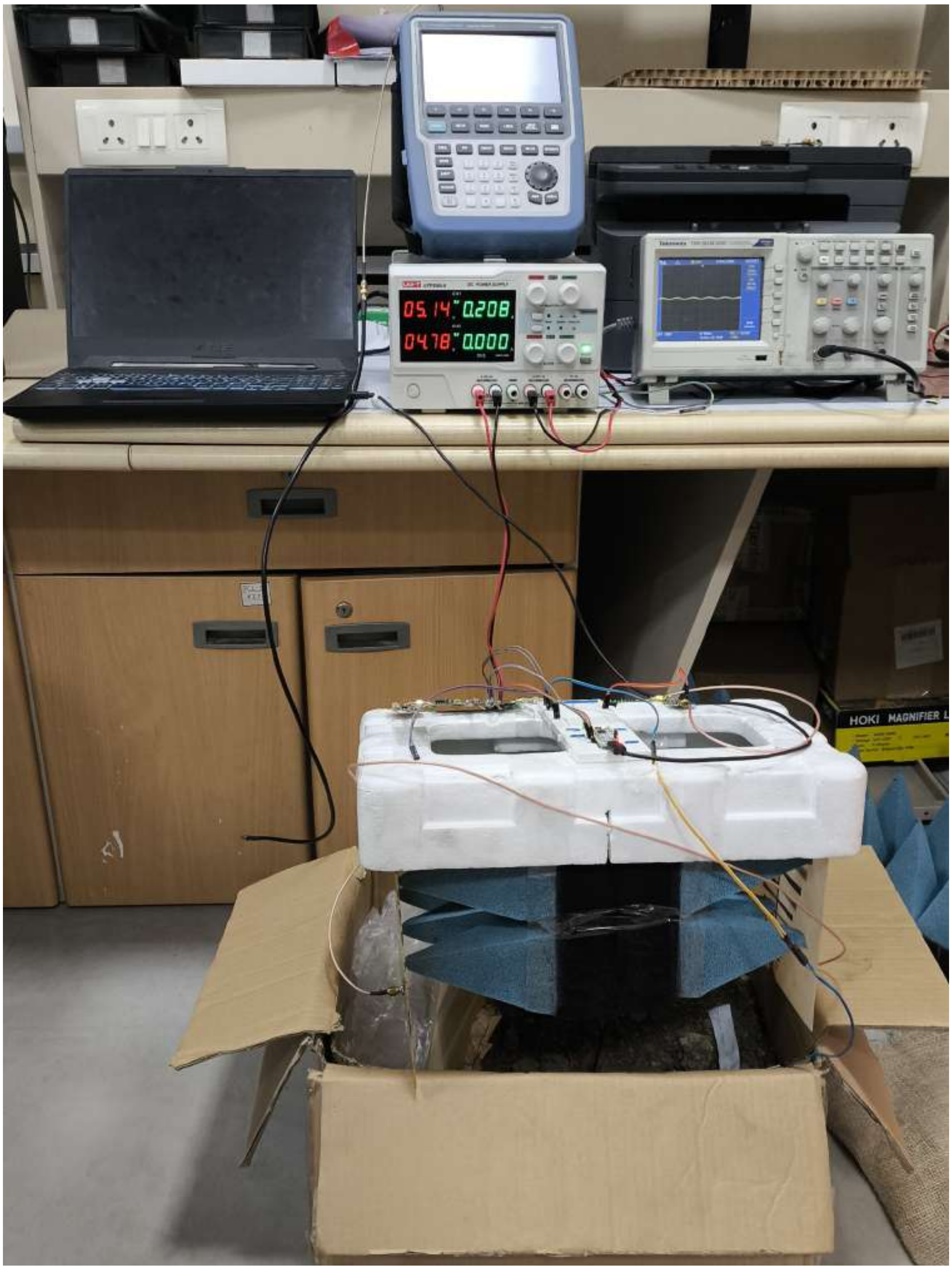


Fig. 5.1: Experimental setup showing antennas, integrated PCB and data acquisition interface.

5.2 Results for Detection of Various Targets in Air

This section presents the system's performance when different targets were placed in free space directly in front of the antenna pair. Testing in air provides a baseline response without the effects of soil permittivity or attenuation.

5.2.1 Detection of Metal targets at different distances in Air

Metallic objects, because of their high conductivity, generate strong reflections and serve as a good benchmark for validating the system. Several metal plates and rods of different sizes were tested, and the corresponding amplitude and phase responses are shown in Fig. 5.2. Across all tests, the radar showed a sharp and distinct rise in amplitude whenever a metal object entered the sensing region. The system also demonstrated expected behavior with distance.

At 1 meter, metal targets produced an amplitude variation of about 0.08 V. At 0.6 meters, the variation increased to around 0.10 V, confirming stronger returns at shorter ranges. Larger metal objects consistently produced stronger reflections, reinforcing the linearity and reliability of the detection mechanism.

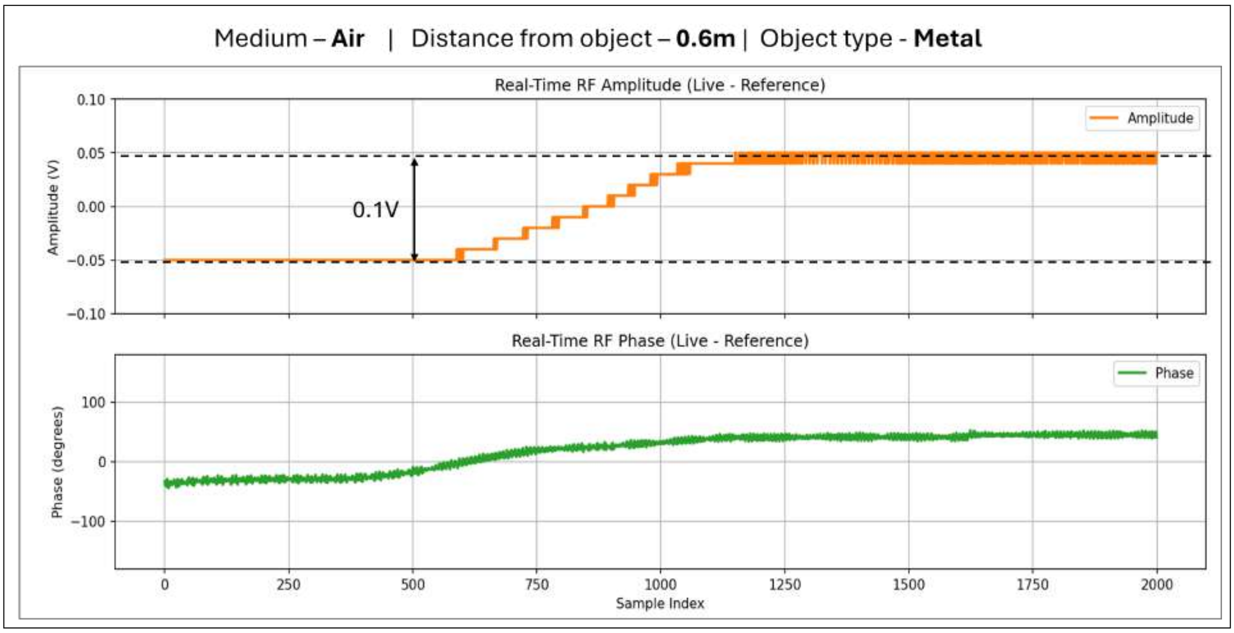


Fig. 5.2: Measured amplitude and phase response for metal target tested in air

5.2.2 Detection of Non-metal targets at different distances in Air

Non-metallic objects such as plastic, ceramic, and wood were also tested to study the system's sensitivity to low-dielectric materials. As expected, these targets produced lower amplitude variations compared to the metals, but the system still responded consistently.

At 0.6 meters, non-metal targets produced approximately 0.05 V of amplitude variation. Although phase changes were less sensitive to target size than to amplitude, they remained useful for identifying smaller low-contrast objects. The results are shown in Fig. 5.3.

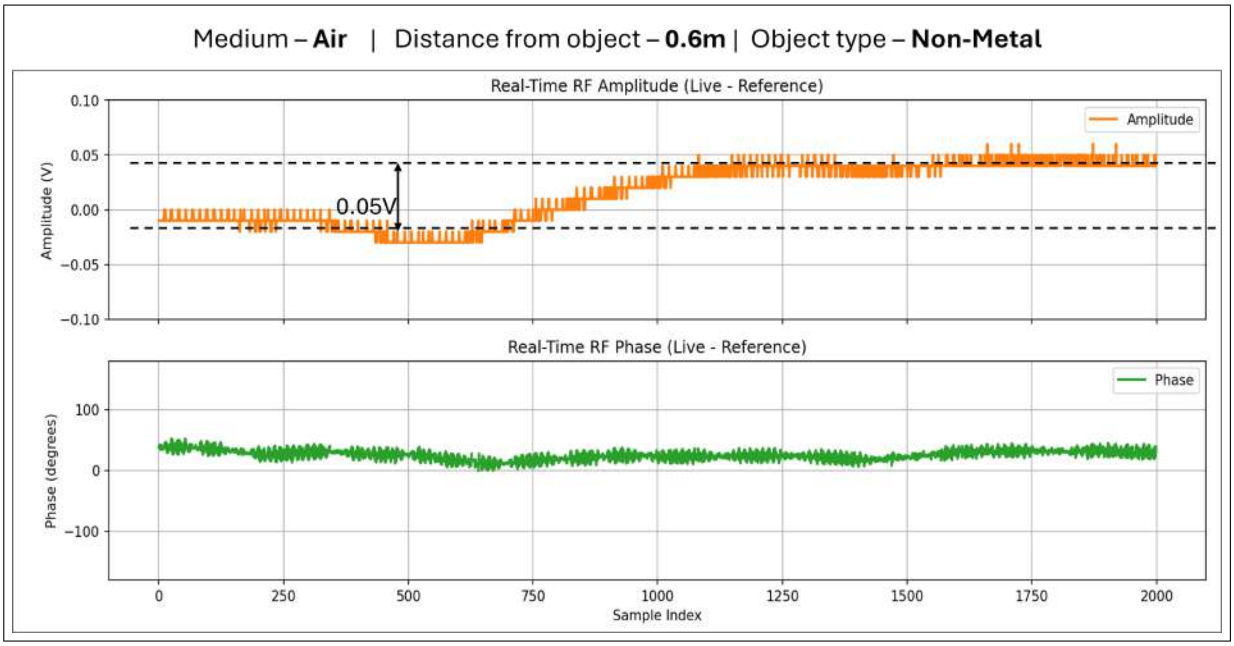


Fig. 5.3: Amplitude and phase response for various non-metal targets tested in air.

5.3 Results for Detection of Various Targets in Soil

To evaluate real-world detection capability, the GPR system was further tested with multiple targets buried in soil. The higher dielectric constant and attenuation of soil reduce reflection strength and alter the beat frequency, making soil-based tests more challenging than air measurements.

5.3.1 Detection of Metal targets at different distances in Soil

Metal objects were buried at different depths under controlled soil conditions. Despite increased losses, the system successfully detected all metal targets, as shown in Fig. 5.4. Both amplitude and phase showed clear deviations whenever a metal object was present.

At 0.6 meters depth, metal targets produced an amplitude variation of about 0.05 V, significantly lower than the 0.10 V observed at the same distance in air due to soil attenuation. Larger targets produced stronger reflections, while deeper targets naturally showed reduced amplitudes.

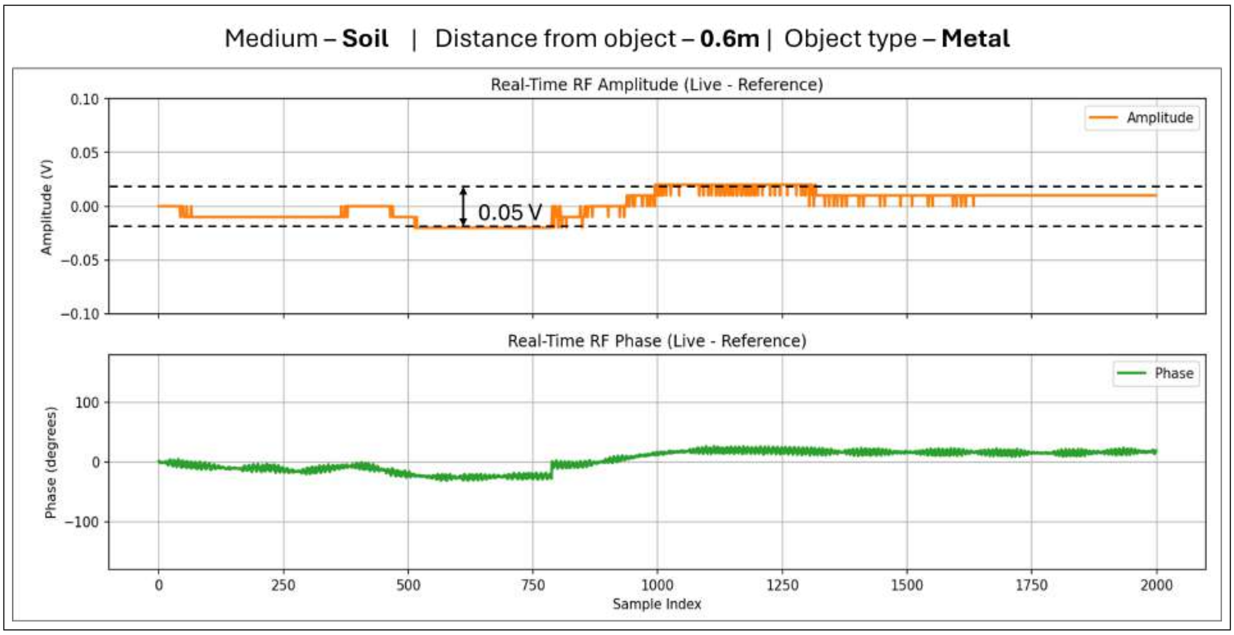


Fig. 5.4: Measured response for metal targets of different sizes buried in soil.

5.3.2 Detection of Non-metal targets at different distances in Soil

Non-metallic buried objects, such as plastic, ceramic, and wood, were also tested to determine the lower limits of detectability. The results in Fig. 5.5 show that even though these targets offer weaker contrast, the system still recorded measurable variations.

At 0.6 meters depth, non-metal targets produced around 0.03 V amplitude variation, which is lower than the 0.05 V observed in air at the same distance. While performance decreases with depth due to increased attenuation, both amplitude and phase changes remained sufficient to indicate target presence.

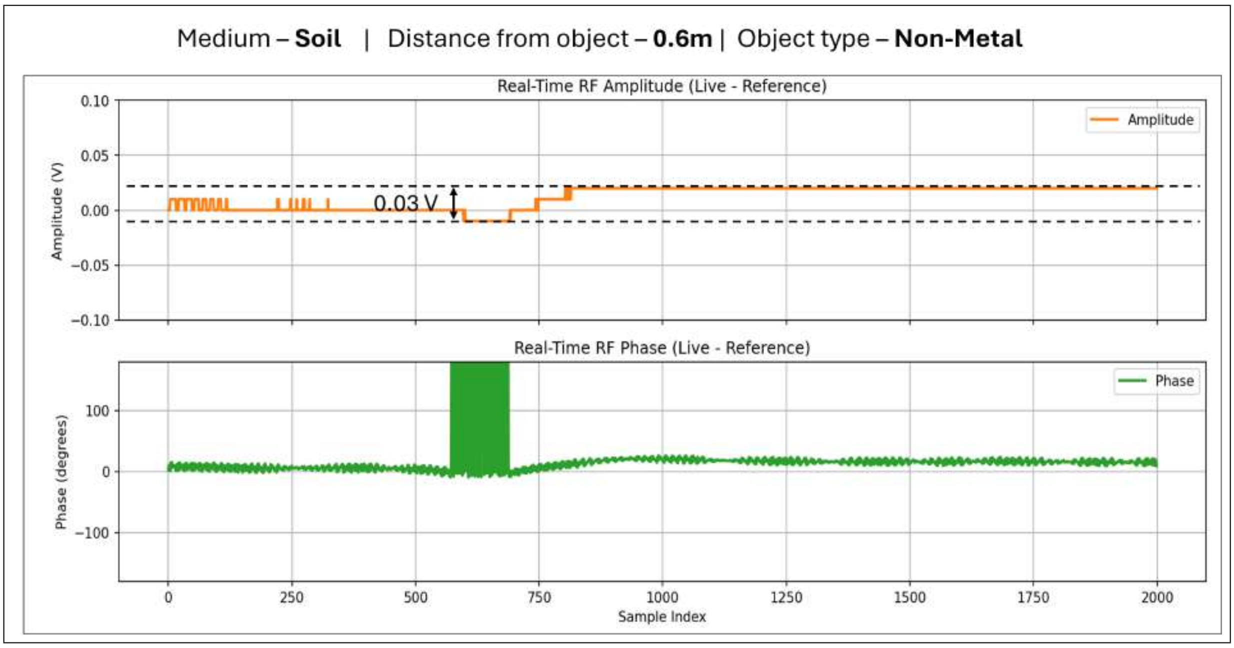


Fig. 5.5: Amplitude and phase response for non-metal targets buried inside soil.

5.4 FFT Calculation and Results

To verify the beat–frequency generated by the FMCW radar, the triangular–wave chirp characteristics were analyzed, and the beat frequency was computed using the theoretical sweep parameters. The corresponding FFT of the received baseband signal, measured using a spectrum analyzer, was used to validate the calculated value.

5.4.1 Theoretical Beat–Frequency Computation

The triangular sweep generated by the TWG operates at

$$f_{\text{tri}} = 9.7 \text{ kHz}.$$

The period of the triangular waveform is therefore

$$T_{\text{tri}} = \frac{1}{f_{\text{tri}}} = \frac{1}{9700} \approx 1.03 \times 10^{-4} \text{ s} = 103 \ \mu\text{s}.$$

Since the chirp uses only half of the triangular cycle, the effective chirp duration becomes

$$T_{\text{chirp}} = \frac{T_{\text{tri}}}{2} = \frac{1}{2f_{\text{tri}}} = \frac{1}{2 \times 9700} \approx 5.15 \times 10^{-5} \text{ s} = 51.5 \ \mu\text{s}.$$

The system sweeps from $f_{\text{min}} = 610$ MHz to $f_{\text{max}} = 790$ MHz, giving a bandwidth of

$$B = f_{\max} - f_{\min} = 790 - 610 = 180 \text{ MHz} = 180 \times 10^6 \text{ Hz}.$$

The sweep slope is therefore

$$S = \frac{B}{T_{\text{chirp}}} = \frac{180 \times 10^6}{51.5 \times 10^{-6}} \approx 3.49 \times 10^{12} \text{ Hz/s}.$$

Using the standard FMCW relation for beat frequency,

$$f_b = \frac{2RS}{c},$$

with $R = 0.6$ m (target distance) and $c = 3 \times 10^8$ m/s (in air),

$$f_b = \frac{2 \times 0.6 \times 3.49 \times 10^{12}}{3 \times 10^8} = 1.3968 \times 10^4 \text{ Hz} \approx 13.97 \text{ kHz}.$$

Thus, the expected beat frequency for a target at 60 cm is

$$\boxed{f_b \approx 14 \text{ kHz}}.$$

5.4.2 Measured FFT Result

The baseband output of the IQ demodulator was observed and the FFT spectrum in Fig. 5.6 shows a dominant peak at 12.7 kHz, which is in close agreement with the theoretical estimation of 14 kHz. Minor deviations may arise due to VCO nonlinearity and propagation characteristics of the surrounding medium.

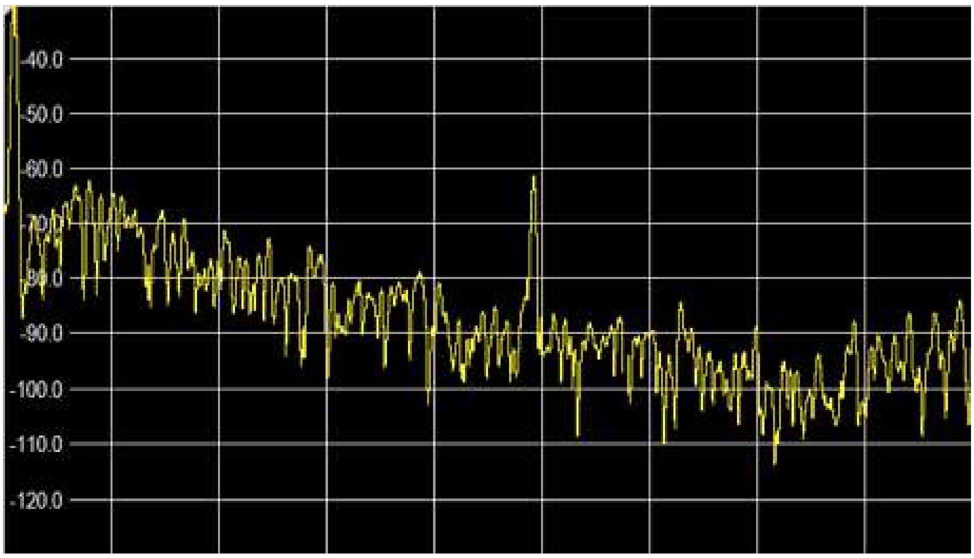


Fig. 5.6: Measured FFT spectrum showing beat-frequency peak at 12.7 kHz.

Chapter 6

Conclusions and Future Work

6.1 Conclusions

In this work, all the RF building blocks required for a low–cost FMCW Ground Penetrating Radar (GPR) transceiver were successfully designed, fabricated, and tested. The developed modules include the VCO-based chirp generator, RF amplification stages, IQ demodulator, baseband signal conditioning network, and the digital processing framework using an STM32 microcontroller.

The complete chain was experimentally verified using controlled measurements, and the system was able to generate the required frequency sweep, maintain stable gain, and produce clean I/Q baseband outputs suitable for further processing. The results confirm that the proposed setup meets the performance requirements for short–range subsurface sensing and landmine–detection scenarios. Overall, the prototype shows strong signal integrity, stable performance, and consistent results across all test scenarios.

6.2 Future Scope of Work

6.2.1 System-Level Integration and Field Deployment

While each RF module has been tested successfully, the current setup uses multiple individual PCBs. A key next step is to consolidate all circuits into a single, compact multilayer PCB. This would improve mechanical robustness, reduce RF losses, and make the system easier to deploy. In addition, full-scale field tests in outdoor environments—with different soil types, moisture levels, and burial depths—will help assess how the system performs under real operating conditions.

6.2.2 Extending Detection Range and Penetration Depth

Right now, the penetration capability of the system is mainly constrained by antenna bandwidth, transmit power, and soil attenuation. Future work could focus on refining the RF front-end, improving power efficiency, and experimenting with antenna structures designed for deeper ground penetration. Systematic testing across various soil conditions will also provide clearer insights into performance limits and guide further design changes.

6.2.3 Signal Processing Enhancements

6.2.3.1 Generating Pseudo-Images of Buried Objects

The current system extracts amplitude and phase information from the received echoes, but this data is not yet converted into any form of image. A natural next step is to develop a 2D reconstruction method that can generate pseudo-images of subsurface objects. This may include adding raw-data interpolation, range migration processing, or a simplified B-scan obtained from multiple linear radar sweeps.

6.2.3.2 Integrating ML Algorithms for More Accurate Detection

Machine learning techniques can significantly improve the identification of buried targets. By building a dataset of I/Q measurements from different types of objects, ML models can be trained to classify metallic and non-metallic targets, estimate depth, and suppress clutter more effectively. Embedding such models in the microcontroller or offloading them to an external processor would enhance the system's real-time detection and decision-making capabilities.

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Appendix A

Signal Processing Codes

A.1 STM32 – Codes for Extracting I/Q Data and Computing Magnitude and Phase

```
/* Includes*/
#include "main.h"

/* Private includes */
/* USER CODE BEGIN Includes */
#include "arm_math.h"
/* USER CODE END Includes */

/* Private define */
#define buffer_size 500
#define uart_baud_rate 1000000

/* Private variables */
ADC_HandleTypeDef hadc1;
ADC_HandleTypeDef hadc2;
DMA_HandleTypeDef hdma_adc1;
DMA_HandleTypeDef hdma_adc2;

CORDIC_HandleTypeDef hcordic;
FMAC_HandleTypeDef hfmac;
UART_HandleTypeDef hlpuart1;

TIM_HandleTypeDef htim6;
TIM_HandleTypeDef htim7;
```

```
TIM_HandleTypeDef htim16;

/* USER CODE BEGIN PV */
int adc1_buffer[buffer_size];
int adc2_buffer[buffer_size];

float conv_3v3 = 3.3/2048;
int call_count1 = 0, call_count2 = 0;

volatile float32_t time_taken3, time_taken4;
volatile float32_t adc1_sample_rate, adc2_sample_rate;

volatile float32_t rf_amp, rf_phase;
volatile float32_t rf_amp_final, rf_phase_final;
/* USER CODE END PV */

/* Callback for ADC DMA */
void HAL_ADC_ConvCpltCallback(ADC_HandleTypeDef *hadc)
{
    if (hadc->Instance == ADC1) {
        time_taken3 = __HAL_TIM_GET_COUNTER(&htim6);
        call_count1++;
        adc1_sample_rate = buffer_size / time_taken3;
    }
    else if (hadc->Instance == ADC2) {
        time_taken4 = __HAL_TIM_GET_COUNTER(&htim16);
        call_count2++;
        adc2_sample_rate = buffer_size / time_taken4;
    }

    if (call_count1 == 1 && call_count2 == 1)
    {
        HAL_TIM_Base_Stop(&htim7);

        rf_amp = 0.0f;
        rf_phase = 0.0f;
```

```
float sum_cos = 0.0f;
float sum_sin = 0.0f;

for (int i = 0; i < buffer_size; i++)
{
    float adc1_value_f = (adc1_buffer[i] - 2047) * conv_3v3;
    float adc2_value_f = (adc2_buffer[i] - 2047) * conv_3v3;

    float mag;
    arm_sqrt_f32(adc1_value_f * adc1_value_f +
                 adc2_value_f * adc2_value_f, &mag);

    rf_amp += mag;
    sum_cos += adc2_value_f / mag;
    sum_sin += adc1_value_f / mag;
}

rf_amp_final = rf_amp / buffer_size;
rf_phase_final = atan2f(sum_sin / buffer_size,
                        sum_cos / buffer_size) * 180 / 3.14142;

char buffer[20];
sprintf(buffer, "A:%.2f\n", rf_amp_final);
HAL_UART_Transmit(&hlpuart1, (uint8_t*)buffer, strlen(buffer),
↪  HAL_MAX_DELAY);

sprintf(buffer, "P:%.2f\n", rf_phase_final);
HAL_UART_Transmit(&hlpuart1, (uint8_t*)buffer, strlen(buffer),
↪  HAL_MAX_DELAY);

HAL_TIM_Base_Start(&htim7);
htim16.Instance->CNT = 0;
htim6.Instance->CNT = 0;
```

```
        call_count1 = 0;
        call_count2 = 0;
    }
}

int main(void)
{
    HAL_Init();
    SystemClock_Config();

    MX_GPIO_Init();
    MX_DMA_Init();
    MX_ADC1_Init();
    MX_TIM6_Init();
    MX_TIM7_Init();
    MX_ADC2_Init();
    MX_TIM16_Init();
    MX_CORDIC_Init();
    MX_FMAC_Init();
    MX_LPUART1_UART_Init();

    HAL_ADCEx_Calibration_Start(&hadc1, ADC_DIFFERENTIAL_ENDED);
    HAL_ADCEx_Calibration_Start(&hadc2, ADC_DIFFERENTIAL_ENDED);

    HAL_TIM_Base_Start(&htim6);
    HAL_TIM_Base_Start(&htim16);

    HAL_ADC_Start_DMA(&hadc1, adc1_buffer, buffer_size);
    HAL_ADC_Start_DMA(&hadc2, adc2_buffer, buffer_size);

    HAL_TIM_Base_Start(&htim7);

    while (1) {
        /* Main loop intentionally empty */
    }
}
```

A.2 Python - Codes for displaying data received from STM 32 in PC

```
import serial
import numpy as np
import matplotlib.pyplot as plt
import time
from collections import deque
from matplotlib.widgets import Button

# === USER CONFIG ===
port = 'COM5'
baudrate = 1000000
num_samples = 2000
draw_interval = 0.02
y_limits_amp = [-0.1, 0.1]
y_limits_phase = [-180, 180]
smoothing_window = 1

# ====================
ser = serial.Serial(port, baudrate, timeout=0)
time.sleep(1)
print("Reading amplitude and phase data from UART...")

# Buffers
amp_buffer = np.zeros(num_samples, dtype=float)
phase_buffer = np.zeros(num_samples, dtype=float)

# Reference buffers (initialized to zero)
amp_reference = np.zeros(num_samples, dtype=float)
phase_reference = np.zeros(num_samples, dtype=float)
write_idx = 0
amp_window = deque(maxlen=smoothing_window)
phase_window = deque(maxlen=smoothing_window)
```

```
# Plot setup
plt.ion()
fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(12, 8), sharex=True)
line_amp, = ax1.plot(np.arange(num_samples), amp_buffer, color='C1',
↪  linewidth=2, label='Amplitude')
ax1.set_title("Real-Time RF Amplitude (Live - Reference)")
ax1.set_ylabel("Amplitude (V)")
ax1.grid(True)
ax1.legend(loc='upper right')
ax1.set_ylim(y_limits_amp)

line_phase, = ax2.plot(np.arange(num_samples), phase_buffer, color='C2',
↪  linewidth=2, label='Phase')
ax2.set_title("Real-Time RF Phase (Live - Reference)")
ax2.set_xlabel("Sample Index")
ax2.set_ylabel("Phase (degrees)")
ax2.grid(True)
ax2.legend(loc='upper right')
ax2.set_ylim(y_limits_phase)
plt.tight_layout(rect=[0, 0.1, 1, 1])

# === BUTTON ===
def save_reference(event):
    global amp_reference, phase_reference
    amp_reference = amp_buffer.copy()
    phase_reference = phase_buffer.copy()
    print("Reference saved!")

ax_button = plt.axes([0.4, 0.02, 0.2, 0.05])
btn_ref = Button(ax_button, "Save Reference", color='lightgray')
btn_ref.on_clicked(save_reference)

plt.show(block=False)

last_draw = time.time()
```

```
buffer_line = b""

try:
    while True:
        if ser.in_waiting:
            data = ser.read(ser.in_waiting)
            buffer_line += data
            lines = buffer_line.split(b'\n')
            buffer_line = lines[-1]  # leftover partial line

            for raw in lines[:-1]:
                try:
                    line = raw.decode().strip()
                    if line.startswith("A:"):
                        amp_val = float(line.split(":")[1])
                        amp_window.append(amp_val)
                        amp_buffer[write_idx] = np.mean(amp_window)

                    elif line.startswith("P:"):
                        phase_val = float(line.split(":")[1])
                        phase_window.append(phase_val)
                        phase_buffer[write_idx] = np.mean(phase_window)
                        write_idx = (write_idx + 1) % num_samples

                except Exception:
                    continue

        if time.time() - last_draw >= draw_interval:
            # Live - Reference
            amp_diff = amp_buffer - amp_reference
            phase_diff = phase_buffer - phase_reference
            line_amp.set_ydata(np.roll(amp_diff, -write_idx))
            line_phase.set_ydata(np.roll(phase_diff, -write_idx))

            fig.canvas.draw_idle()
            fig.canvas.flush_events()
```